Sustaining Coordination in a Dynamic Stag-Hunt Game*

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Abstract

Many collective goals are secured only after sustained coordination, and a single failure can erase accumulated progress. In my dynamic stag-hunt game, consecutive successful coordination advances a public state toward a protection threshold; any failure resets it, and an outside shock before the threshold imposes a loss. I characterize Markov Perfect Equilibria and develop a state-dependent dynamic basin of attraction measuring how much strategic uncertainty agents can tolerate while coordination remains optimal. In an experiment varying group size and the safe payoff and eliciting choice probabilities, subjects choose the risky action more as groups approach the threshold, and coordination persists beyond it. A lower safe payoff increases coordination; smaller groups coordinate more only when the safe payoff is high. Post-failure behavior differs from initial behavior in only two of the four treatments, and in opposite directions. Individual mixing is common initially and fades with progress and experience.

Keywords: *Coordination, Markov Perfect Equilibrium, Dynamic Basin of Attraction*

JEL Classification: *C73, C92, D91*

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1 Introduction

Many coordination problems are not resolved in a single interaction. A collective goal is often secured only after a group coordinates successfully over a sustained period: each success brings the group closer to the goal, while a single breakdown can erase the accumulated progress. Communities fighting an infectious disease, for example, can eliminate it only if transmission is halted everywhere for a sustained period, and a lapse in any one community can restart the effort. Teams adopting a new practice, such as a safety protocol or a new technology, benefit only if every member follows it, and an early lapse can send the team back to old habits before the practice becomes routine. Minority communities seeking to preserve their language or culture face a similar challenge: preservation requires costly effort from many members, assimilation offers a safer alternative, and progress can be lost when participation breaks down.[1] In each case, the effort is also exposed to outside shocks, such as a new importation of the disease, a change in leadership, or a wave of out-migration, that can end it before the goal is secured. The value of doing one's part today therefore depends on how much progress the group has already made.

The stag hunt is the canonical model of such coordination problems. Each player chooses between a risky action, which pays off only if others also choose it, and a safe action, which guarantees a lower payoff regardless of others' choices. Both universal risky actions and universal safe actions are equilibria, and which one players select depends on how much strategic uncertainty, that is, uncertainty about whether others will do their part, they can tolerate (Harsanyi and Selten, 1988; Cooper et al., 1990; Van Huyck et al., 1990). A standard measure of this tolerance is the size of the *basin of attraction* of the risky action: the largest probability that the other player chooses the safe action at which the risky action remains a best response. The concept originates in evolutionary game theory (Kandori et al., 1993; Young, 1993; Ellison, 1993) and was brought to experimental economics by Dal Bó and Fréchette (2011) in repeated prisoner's dilemmas; Dal Bó et al. (2021) show that it predicts efficient play in one-shot stag hunts.[2] Existing basin measures, however, assign a single number to each game, computed either for a one-shot game or for a repeated game whose stage game never changes.

In the coordination problems above, the incentive to coordinate is not fixed. Because each success moves the group closer to its goal, the amount of strategic uncertainty a member

[1] Related economic models have studied incentives for minority assimilation and language acquisition, network externalities in language use, and the intergenerational transmission of cultural traits; see Church and King (1993); Lazear (1999); Bisin and Verdier (2000).

[2] Related experimental work studies optimization incentives, risk and payoff dominance, relative riskiness, and history dependence in coordination games (Battalio et al., 2001; Schmidt et al., 2003; Dubois et al., 2012; Kendall, 2022; Jagau, 2024; Silva, 2024).

can tolerate while still doing their part changes as progress accumulates, and a single static measure cannot describe how the robustness of coordination evolves over the course of a group's effort. The dynamic setting also raises a question that one-shot games cannot pose. Because the same level of progress can be reached through different histories, players may condition their choices only on current progress (state-dependent behavior) or also on how the group got there, for example by giving up after a failure (history-dependent behavior). This paper asks two questions. **First, how robust is coordination at each stage of progress, and how does this robustness depend on group size and the attractiveness of the safe action? Second, does behavior depend on progress alone, or also on past failures?**

I study these questions in a dynamic stag-hunt game with three key features. First, coordination in each round is fragile: an agent who chooses the risky action X receives the benefit only if all other group members also choose X, whereas the safe action Y guarantees a lower payoff regardless of the group outcome. Second, progress accumulates but can be lost: a publicly observed state k counts consecutive all-X rounds, in which every group member chooses X, and a single Y choice resets k to zero. Third, progress matters because of an outside shock: at the end of each round, the game ends with a constant probability, and if it ends before k reaches a threshold, every member's final-round payoff is overwritten to a loss. Once k reaches the threshold, the group is protected. In terms of the examples above, k measures how long the group has sustained its effort, a reset is a lapse, and the shock is an outside disruption. Because the stage payoffs are the same in every state, the state affects incentives only through continuation values: the closer the group is to the threshold, the fewer all-X rounds it still needs to secure protection, and the more progress a failure destroys.

I analyze the game in two complementary ways. First, I characterize its Markov Perfect Equilibria (MPE), in which each agent's action depends only on the current public state k, independent of the history (Maskin and Tirole, 1988a,b, 2001; Dutta, 1995). Every symmetric pure Markov strategy is an MPE, so the game has many equilibria. I focus on *monotone* equilibria, in which the probability of choosing X is weakly increasing in k. Along the path of play, the monotone pure MPEs predict either full coordination on X or permanent coordination on Y. For thresholds of one to three rounds, the unique on-path monotone mixed MPE, which serves as the benchmark for the experiment, prescribes mixing at $k = 0$ and choosing X for sure once the group has achieved one all-X round, so strategic uncertainty is concentrated at the initial state. As is typical of mixed equilibria in coordination games, however, its mixing probability has counterintuitive comparative statics: it rises with the safe payoff and with group size.

Second, to obtain predictions on how the robustness of coordination varies with progress and with the parameters of the game, I define a *dynamic basin of attraction*. Following Dal Bó and Fréchette (2011), I compare Grim Trigger, under which an agent chooses X as long as every previous round has been all-X and switches permanently to Y after the first round that is not, with All-Y. Because coordination requires every other group member to choose X, I measure the basin per opponent (Boczoń et al., 2025): the dynamic basin at state k is the largest probability that each other member plays All-Y such that Grim Trigger remains a best response. The dynamic basin has three properties that the experiment tests. It expands as the group makes progress, but the expansion stops once protection is one round away; it is larger when the safe action pays less; and it is larger when the group is smaller. At the initial state, the dynamic basin coincides with the belief cutoff implied by the monotone mixed MPE, so the Markov and Grim Trigger approaches deliver the same measure of strategic uncertainty at the state where that uncertainty is greatest. Finally, the two approaches yield a direct test of state versus history dependence: under a Markov strategy, an agent's behavior at $k = 0$ should be the same whether the game has just started or the state has just been reset, whereas Grim Trigger prescribes a permanent switch to Y after a reset.

I test these predictions in a preregistered laboratory experiment with 164 subjects in 12 sessions. In every match, the threshold is two consecutive all-X rounds, the match continues after each round with probability 0.8, and if it ends before the group reaches the threshold, every member's final-round payoff is overwritten to -100 points. An all-X round pays each member 100 points, choosing X otherwise pays 0 points, and choosing Y pays a safe payoff of 15 or 85 points ($u_Y \in \{0.15, 0.85\}$ in normalized terms). Group size varies between subjects ($n \in \{2, 4\}$) and the safe payoff varies within subjects, yielding four treatment conditions, $2l$, $2h$, $4l$, and $4h$, where the number denotes group size and l (low) and h (high) denote the safe payoff. The treatments generate exogenous variation in the dynamic basin: larger groups require each agent to hold more optimistic beliefs about each opponent, and a higher safe payoff makes Y more attractive. The dynamic basin is close to one at all states in $2l$ but below 0.5 at all states in $4h$. In every condition, it expands from $k = 0$ to $k = 1$ but not from $k = 1$ to $k = 2$, so the probability assigned to X should increase more in the first step than in the second. In each round, subjects assign a probability to X, and the computer implements X or Y according to that probability (Romero and Rosokha, 2023). This probability elicitation distinguishes individual mixing from population heterogeneity, in which some subjects choose X and others choose Y, and allows me to compare observed behavior with the mixed MPE benchmark. Two additional baseline sessions with $n = 4$ remove the public state and the terminal loss and provide a descriptive benchmark for a

standard repeated stag hunt.

The experiment yields four main findings. First, coordination depends systematically on progress. In the pooled sample, the probability assigned to X increases as the group moves toward the threshold, with the largest increase occurring between $k = 0$ and $k = 1$, and coordination is highly persistent once groups reach the threshold. The predicted gap between the first and second increments has the expected sign but falls just short of statistical significance at the 5% level. Because the state is generated by past play, I interpret these state gradients as conditional associations rather than causal effects. In line with the monotone mixed MPE, interior probabilities are concentrated at $k = 0$ and behavior is close to pure X at higher states, so the aggregate pattern resembles, although it does not exactly implement, that benchmark. Second, on state versus history dependence, I do not find a statistically significant difference in the pooled sample between behavior at the initial $k = 0$ state and behavior after a reset. The treatment-specific estimates, however, reveal significant history dependence in opposite directions: post-reset probabilities assigned to X are 8.9 percentage points higher in $2l$, the condition with the largest dynamic basin, and 22.6 percentage points lower in $4h$, the condition with the smallest. A coordination failure thus appears to discourage coordination where strategic uncertainty is greatest. In no treatment, however, does post-reset behavior resemble the permanent switch to Y prescribed by Grim Trigger. Third, on group size and the safe payoff, and consistent with the comparative statics of the dynamic basin, lowering the safe payoff increases the probability assigned to X for both group sizes. Group size matters when the safe payoff is high, where groups of two assign substantially higher probabilities to X than groups of four, but not when the safe payoff is low, where coordination is already near the upper bound in both group sizes. Fourth, probability elicitation reveals that interior choice probabilities are common, particularly at the initial state. Mixing becomes substantially less frequent at higher states, where pure-X choices dominate, and it also declines with experience over repeated matches. Finally, relative to the baseline sessions, the dynamic structure is associated with more X choices when the safe payoff is high, and the difference is concentrated at $k = 0$. Together, these results show that the robustness of coordination depends systematically on accumulated progress, as the dynamic basin predicts, although the current state does not fully summarize behavior.

This paper contributes to the literature on strategic uncertainty and equilibrium selection by extending basin-of-attraction reasoning to a game in which successful coordination itself changes the state. Three closely related papers study basin measures in dynamic settings. Vespa and Wilson (2016) use basin-of-attraction reasoning to compare competing equilibrium predictions in a dynamic game; Gill et al. (2026) develop basin measures for equilibrium

selection over investment and task choice decisions in dynamic team skill formation; and Rosokha and Wei (2024) relate basin-of-attraction reasoning to equilibrium selection among history-contingent, state-contingent Markov, and state-and-history-contingent strategies in a dynamic prisoner's dilemma. None considers a dynamic stag hunt in which successful coordination advances a publicly observed state toward a threshold. My dynamic basin also adopts the per-opponent formulation that Boczoń et al. (2025) develop for N-player repeated games, although their basin is not state-dependent. The dynamic basin developed here delivers state-by-state predictions and comparative statics in group size and the safe payoff, and it coincides with the mixed-MPE belief cutoff at the initial state, linking the Markov and history-dependent approaches. The experimental evidence largely supports these predictions.

This paper also relates to experiments on stag-hunt coordination over time (Rankin et al., 2000; Al-Ubaydli et al., 2013; Duffy and Fehr, 2018; Brocas and Carrillo, 2025). In these environments, repeated interaction allows behavior and coordination to evolve, but current coordination does not itself generate the payoff-relevant progress state studied here. Closer to my setting, Gerlagh and van der Heijden (2024) experimentally study a dynamic three-player stag-hunt game in which groups can move over several rounds from one stable state to a higher-payoff stable state, and they compare neutral and environmental framing. Their focus is on transition dynamics and framing. In contrast, this paper studies an endogenous public state generated by successful coordination and equilibrium selection through a state-dependent basin of attraction. My baseline sessions, which remove the public state and the terminal loss, suggest that this structure matters most when coordination is otherwise difficult to sustain.

This paper further relates to experimental work on whether behavior in dynamic games conditions on the payoff-relevant state, as Markov strategies prescribe. Evidence from dynamic public-investment, common-pool, cooperation, and team-production environments shows that subjects often respond systematically to the current state, although their behavior need not coincide exactly with Markov equilibrium predictions (Battaglini et al., 2012, 2016; Agranov et al., 2016; Vespa and Wilson, 2019; Vespa, 2020; Salz and Vespa, 2020; Rosokha and Wei, 2024; Gill et al., 2026). In coordination games, past play and precedents are known to shape subsequent equilibrium selection (Van Huyck et al., 1991; Rankin et al., 2000; Schmidt et al., 2003). In my game, the stage game is identical in every state and a coordination failure returns the game to exactly its initial state, so comparing behavior at the initial $k = 0$ with behavior after a reset holds the payoff-relevant state fixed and provides a clean test of state against history dependence. I find state-dependent behavior in the pooled sample, but history dependence in opposite directions in the $2l$ and $4h$ treatments,

the conditions with the largest and smallest dynamic basins.

Finally, this paper extends the probability elicitation of Romero and Rosokha (2023) from the indefinitely repeated prisoner's dilemma to dynamic coordination. It shows that individual mixing is a common feature of behavior that is concentrated at the initial state and declines with both progress and experience, so that heterogeneity in observed actions cannot be attributed exclusively to heterogeneity in pure decisions. The elicited probabilities also allow a direct comparison with the state-contingent mixing pattern of the mixed MPE.

The paper is organized as follows. Section 2 presents the model, characterizes the Markov Perfect Equilibria, and develops the dynamic basin of attraction. Section 3 describes the experimental design and hypotheses, Section 4 presents the results, and Section 5 discusses and concludes.

2 Theoretical Framework

2.1 Model Settings

The model is an n-player dynamic stag-hunt game with a public state that records consecutive all-X coordination and an exogenous risk shock. In each round, agent $i \in \{1, \ldots, n\}$ chooses between two actions: X and Y. To normalize the payoffs, let the payoff of action X be 1 if the group outcome is all-X (i.e., every group member chooses X), and 0 otherwise; let the payoff of action Y be $u_Y \in (0, 1)$. See Table 1 for the stage payoffs of Agent i.

		Agent j	
		X	Y
Agent i	X	$(1, 1)$	$(0, u_Y)$
	Y	$(u_Y, 0)$	(u_Y, u_Y)

(a) Group size $n = 2$

		Group outcome	
		all-X	not all-X
Agent i	X	1	0
	Y		u_Y

(b) General group size n

Table 1: Normalized Stage Payoffs

An outside shock occurs with probability $1 - \delta \in (0, 1)$ at the end of each round, after all group members have made their choices, where δ is the continuation probability of this game. Let $t \leq T$ be the round number, where T marks the last round. Let k_t denote the public state at the end of round t. The state is a capped counter of consecutive all-X rounds. Given $k_0 = 0$, the counter k_t updates as

$$k_t = \begin{cases} \min\{k_{t-1} + 1, \bar{K}\} & \text{if all-}X\text{ in round } t, \\ 0, & \text{otherwise} \end{cases}$$

where $\bar{K}$ is the state threshold. The group is *protected* once k reaches $\bar{K}$. Thus, at the beginning of round t, agents observe the inherited state k_{t-1} before making their decisions. When the time index is unnecessary, I use k to denote the state observed at the beginning of a decision round. When the shock is realized (in round T) and $k_T < \bar{K}$, the payoffs in round T are overwritten to $-d$ for all members. Otherwise, payoffs are not overwritten. Although the within-round stage payoffs are identical across states, the public state k is payoff-relevant through continuation values: if the game ends before the state reaches $\bar{K}$, every member's final-round payoff is overwritten to $-d$. Hence, the environment is a dynamic stochastic game rather than a standard indefinitely repeated game with a fixed stage game.

In a standard repeated stag hunt, the same stage game is repeated over time, but current actions do not typically change an endogenous payoff-relevant state of the game. In the

present environment, by contrast, coordination on X affects both current payoffs and the public state k. The current state therefore summarizes the group's progress toward protection and directly affects continuation incentives. At the same time, the environment is stationary because the continuation probability δ is constant. Conditional on the same current state k, the continuation problem is therefore the same regardless of how many rounds have already elapsed or how that state was reached. In particular, after any reset to $k = 0$, the incentives to begin coordinating toward the threshold are the same as at any other visit to $k = 0$.

Before characterizing equilibrium behavior, I establish the socially efficient benchmark. At any state k, unanimous choice of X yields a total stage payoff of n, whereas any action profile containing at least one Y yields a total stage payoff no greater than $nu_Y < n$. Moreover, when $k < \bar{K}$, unanimous X moves the group toward protection, while any outcome containing Y resets the state to zero. Thus, unanimous X weakly improves both the current group payoff and the continuation state.

Remark. At every state $k \in \{0, \ldots, \bar{K}\}$, it is socially efficient for all agents to choose X. Consequently, the socially efficient path reaches the protected state after $\bar{K}$ consecutive all-X rounds.

Importantly, the socially efficient action is the same in all experimental treatments. Changes in u_Y and n therefore affect the strategic difficulty of implementing the efficient outcome rather than the efficient benchmark itself. The equilibrium analysis below studies how this difficulty varies with the current state and treatment parameters.

2.2 Markov Perfect Equilibrium

Because the state k summarizes everything that is payoff-relevant about the past, a natural benchmark is a *Markov strategy*, in which an agent's choice depends only on the current state and not on the detailed history of play. A *Markov Perfect Equilibrium* (MPE) is a profile of Markov strategies in which each agent's strategy is a best response to the others' strategies in every state (Maskin and Tirole, 1988a,b, 2001). It refines Subgame Perfect Equilibrium (SPE), the standard equilibrium concept for dynamic games, which requires best responses after every possible history. I focus on Markov strategies that are *symmetric*, so that all agents use the same strategy, and *stationary*, so that the strategy does not change over time (Dutta, 1995). Such a strategy is summarized by $\sigma = (\sigma_0, \sigma_1, \ldots, \sigma_{\bar{K}})$, where $\sigma_k \in [0, 1]$ is the probability that an agent chooses X in state k. A strategy is *pure* if $\sigma_k \in \{0, 1\}$ in every state and *mixed* otherwise.

Let V_k denote an agent's *continuation value*, the expected total payoff from the beginning of a round in state k onward when all agents follow σ. Let $Q_k(X)$ and $Q_k(Y)$ denote the

expected payoff from choosing X or Y in the current round and following σ thereafter. Choosing Y guarantees that the round is not all-X, so the next state is 0, and the payoff is overwritten to $-d$ if the shock arrives:

$$Q_k(Y) = \delta\big(u_Y + V_0\big) + (1-\delta)(-d).$$

An agent who chooses X obtains an all-X round only if all other $n-1$ members also choose X, which occurs with probability $q_k = \sigma_k^{n-1}$. If $k < \bar{K} - 1$, even an all-X round leaves the group unprotected, whereas if $k \geq \bar{K} - 1$, an all-X round reaches (or keeps) the protected state, so the shock no longer triggers the loss:

$$Q_k(X) = \begin{cases} q_k\Big(\delta\big(1+V_{k+1}\big) + (1-\delta)(-d)\Big) + (1-q_k)\Big(\delta V_0 + (1-\delta)(-d)\Big) & \text{if } k < \bar{K}-1, \\ q_k\Big(\delta\big(1+V_{\bar{K}}\big) + (1-\delta)\Big) + (1-q_k)\Big(\delta V_0 + (1-\delta)(-d)\Big) & \text{if } k \geq \bar{K}-1. \end{cases}$$

A symmetric stationary MPE is a collection $\{V_k, \sigma_k\}_{k=0}^{\bar{K}}$ such that, for each state k,

$$V_k = \max\{Q_k(X), Q_k(Y)\}, \qquad \sigma_k = \begin{cases} 1 & \text{if } Q_k(X) > Q_k(Y), \\ 0 & \text{if } Q_k(X) < Q_k(Y), \\ \in (0,1) & \text{only if } Q_k(X) = Q_k(Y). \end{cases}$$

That is, agents choose X for sure when X is strictly better, Y for sure when Y is strictly better, and mix only when they are indifferent.[3] The game has many MPEs. I focus on *monotone* MPEs, in which the probability of choosing X does not decrease as the group makes progress.

Definition 1 (Monotone Markov strategy). A symmetric stationary Markov strategy σ is *monotone* if $\sigma_0 \leq \sigma_1 \leq \cdots \leq \sigma_{\bar{K}}$.

Throughout the paper, *monotone* means weakly increasing in k. A higher state means that the group needs fewer additional all-X rounds to become protected, so it is difficult to justify behaviorally a strategy that chooses X at a low state but abandons X at a higher one. Section 2.2.3 briefly discusses the non-monotone MPEs that this restriction excludes.

For brevity, I write a strategy as a string that lists σ_k in order of $k = 0, 1, \ldots, \bar{K}$, where 0 denotes $\sigma_k = 0$ (choose Y for sure), 1 denotes $\sigma_k = 1$ (choose X for sure), and m denotes interior mixing, $\sigma_k \in (0,1)$. For example, with $\bar{K} = 2$, $(m11)$ prescribes mixing at $k = 0$

[3] The condition is "only if" rather than "if" because indifference is necessary but not sufficient for mixing. When $Q_k(X) = Q_k(Y)$, every $\sigma_k \in [0,1]$ is a best response, including the pure choices $\sigma_k = 0$ and $\sigma_k = 1$. Thus, an agent who mixes must be indifferent, but an indifferent agent need not mix.

and choosing X for sure at $k = 1$ and $k = 2$. Any MPE with $\sigma_0 = 0$ never leaves the initial state: all members choose Y at $k = 0$, so the state never rises to $k = 1$. I call such an MPE *off-path*. By contrast, I call an MPE *on-path* if $\sigma_0 > 0$, so that equilibrium play reaches higher states with positive probability.

2.2.1 Monotone Pure MPEs

In a symmetric pure strategy, the other members either all choose Y or all choose X in a given state. If they all choose Y, choosing X cannot produce an all-X round, so Y is a best response. If they all choose X, choosing X is a best response, because it yields a higher current payoff ($1 > u_Y$) and advances (or keeps) the state instead of resetting it. Hence, every symmetric pure Markov strategy is an MPE.[4] Among them, the monotone ones take a simple threshold form.

Proposition 1 (Monotone pure MPEs). The symmetric monotone pure MPEs are exactly the threshold strategies

$$a^{(r)}(k) = \begin{cases} Y, & k < r, \\ X, & k \geq r, \end{cases} \qquad r \in \{0, 1, \ldots, \bar{K} + 1\}.$$

Hence, there are exactly $\bar{K} + 2$ symmetric monotone pure MPEs.

Only $r = 0$, under which all members choose X in every state, is on-path. For every $r \geq 1$, members choose Y at $k = 0$, so play never leaves $k = 0$ and is identical to that of the all-Y MPE ($r = \bar{K}+1$), the natural coordination-failure benchmark. Thus, along the path of play, monotone pure MPEs predict either full coordination on X or permanent coordination on Y, and neither says how coordination should vary with the state, the safe payoff, or group size. The mixed MPEs below predict how play varies with the state, and the dynamic basin in Section 2.3 also predicts how it varies with the safe payoff and group size.

2.2.2 Monotone Mixed MPEs

In a mixed MPE, $\sigma_k \in (0, 1)$ in at least one state, which requires $Q_k(X) = Q_k(Y)$ in that state. To find all monotone mixed MPEs, I first list every candidate pattern of 0, m, and 1 that never moves "down" in the order $0 \to m \to 1$ as k increases (e.g., $(0m1)$ but not $(m01)$); I call these *regime-monotone* candidates. I then solve the indifference conditions of each candidate in closed form and keep only the candidates whose solved probabilities satisfy

[4] See Appendix A1.4 for the formal statement and proof, as well as the proof of Proposition 1 below.

$\sigma_0 \leq \sigma_1 \leq \cdots \leq \sigma_{\bar{K}}$. The second step is necessary because a candidate such as (mm) can have $\sigma_0 > \sigma_1$ once the two mixing probabilities are solved.[5]

Proposition 2 (On-path monotone mixed MPEs). For $\bar{K} \in \{1, 2, 3\}$, there is exactly one on-path monotone mixed MPE: $(m1)$ for $\bar{K} = 1$, $(m11)$ for $\bar{K} = 2$, and $(m111)$ for $\bar{K} = 3$. In each case, members mix at $k = 0$ and choose X for sure in every state $k \geq 1$.

Table 2 summarizes the monotone pure and mixed MPEs for $\bar{K} = 1, 2, 3$, and Figure 1 reports the closed-form solutions of the three on-path monotone mixed MPEs. In each of them, strategic uncertainty is concentrated at the initial state. Once the group has achieved one all-X round, all members choose X for sure, so the group reaches the protected state and remains there. The mixed MPE $(m11)$ with $\bar{K} = 2$ serves as the mixed-MPE benchmark for the experiment.

Table 2: Symmetric Monotone MPEs for $\bar{K} = 1, 2, 3$

	Pure		Mixed	
$\bar{K}$	All monotone	On-path	All monotone	On-path
1	(00), (01), (11)	(11)	$(0m)$, $(m1)$	$(m1)$
2	(000), (001), (011), (111)	(111)	$(00m)$, $(0mm)$, $(0m1)$, $(m11)$	$(m11)$
3	(0000), (0001), (0011), (0111), (1111)	(1111)	$(000m)$, $(00mm)$, $(00m1)$, $(0m11)$, $(m111)$	$(m111)$

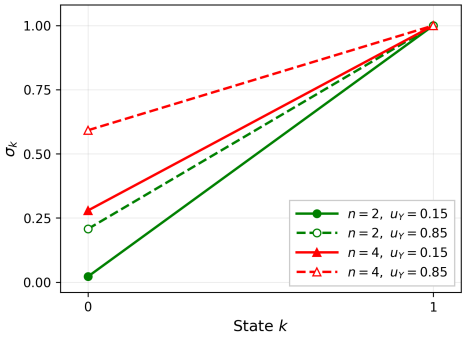
Note: Each profile lists σ_k for $k = 0, 1, \ldots, \bar{K}$. 0 and 1 denote $\sigma_k = 0$ and $\sigma_k = 1$; m denotes interior mixing, $\sigma_k \in (0, 1)$. An MPE is on-path if $\sigma_0 > 0$, so that play can leave the initial state $k = 0$. Appendix Table A1.1 also lists the mixed candidates that fail monotonicity.

The mixing probability σ_0 admits two interpretations. Under the *equilibrium interpretation*, σ_0 is the probability with which each member chooses X at $k = 0$. Under the *best-response interpretation*, σ_0 is a *belief cutoff*: if a member believes that the others choose X for sure at $k \geq 1$ and choose X at $k = 0$ with probability above σ_0, then X is a strict best response at $k = 0$. As with mixed equilibria in coordination games generally, the equilibrium interpretation implies counterintuitive comparative statics (σ_0 rises with u_Y and n), because each member's mixing probability is pinned down by the requirement that the *other* members be indifferent. I adopt the best-response interpretation, under which $1-\sigma_0$ is the largest probability of Y per other member that a member can tolerate while X remains optimal. For example, in $(m11)$ with $\delta = 0.8$ and $d = 1$, $\sigma_0 = 0.02$ when $n = 2$ and $u_Y = 0.15$, whereas $\sigma_0 = 0.62$ when $n = 4$ and $u_Y = 0.85$. Section 2.3 shows that this cutoff coincides with the dynamic basin at the initial state.

[5]Appendix A1 reports the full procedure and the solutions to every candidate for $\bar{K} = 1, 2, 3$.

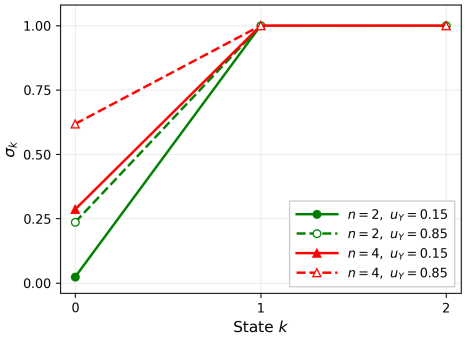
$[\bar{K} = 1$**: MPE** $(m1)]$

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = \frac{1}{1-\delta},$$
$$\sigma_0 = \left(\frac{\delta(1-\delta)u_Y}{1 - \delta^2 u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}}, \qquad \sigma_1 = 1.$$



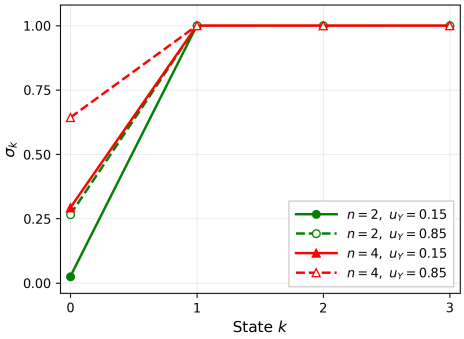
$[\bar{K} = 2$**: MPE** $(m11)]$

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = V_2 = \frac{1}{1-\delta},$$
$$\sigma_0 = \left(\frac{(1-\delta)u_Y}{(2-\delta) - \delta u_Y + (1-\delta)d} \right)^{\frac{1}{n-1}}, \qquad \sigma_1 = \sigma_2 = 1.$$



$[\bar{K} = 3$**: MPE** $(m111)]$

$$V_0 = \frac{\delta}{1-\delta} u_Y - d, \qquad V_1 = \frac{\delta(2-\delta)}{1-\delta} - (1-\delta)d,$$
$$V_2 = V_3 = \frac{1}{1-\delta}, \qquad \sigma_1 = \sigma_2 = \sigma_3 = 1,$$
$$\sigma_0 = \left(\frac{(1-\delta)u_Y}{1 + \delta(1-\delta)(1+d) - \delta u_Y} \right)^{\frac{1}{n-1}}.$$



Notes: The plots illustrate σ_k by state for $\delta = 0.8$ and $d = 1$; green for $n = 2$, red for $n = 4$, solid lines for $u_Y = 0.15$, and dashed lines for $u_Y = 0.85$. Derivations are in Appendix A1; additional parameter comparisons for $(m11)$ are in Appendix A2.

Figure 1: Solutions for the On-path Monotone Mixed MPEs $(m1)$, $(m11)$, and $(m111)$

2.2.3 Non-monotone MPEs

For completeness, the game also has MPEs that are not monotone. Because every symmetric pure Markov strategy is an MPE, there are $2^{\bar{K}+1}$ pure MPEs, of which only the $\bar{K}+2$ in Proposition 1 are monotone. For example, with $\bar{K} = 2$, (100), under which members choose X at $k = 0$ but switch to Y as soon as the group makes progress, is a pure MPE. Some mixed MPEs are also non-monotone. For example, with $\bar{K} = 1$, the mixed MPE (mm) has $\sigma_0 > \sigma_1$.[6] Such strategies abandon X precisely as the group approaches protection. In this

[6]See Appendix A1 for the solution of (mm) and the other non-monotone mixed candidates.

paper, however, I focus only on monotone MPEs.

2.3 History-dependent Strategies and Dynamic Basin of Attraction of Coordination

The game has multiple equilibria, including full coordination on X and permanent coordination on Y. Which one agents select depends on *strategic uncertainty*, that is, uncertainty about whether the other members will do their part. In the experimental literature on indefinitely repeated games, the existence of a cooperative equilibrium alone provides limited guidance for equilibrium selection, whereas measures of robustness to strategic uncertainty provide sharper behavioral predictions (Blonski et al., 2011; Dal Bó and Fréchette, 2011, 2018). This subsection develops such a measure for the present game, in which the robustness of coordination varies with the state.

Static basin of attraction. I begin with the one-shot benchmark of Dal Bó et al. (2021). In a two-player one-shot stag hunt, the size of the *basin of attraction* of the risky action is the largest probability that the other player chooses the safe action such that the risky action remains a best response. With the payoffs in panel (a) of Table 1, if agent i believes that agent j chooses Y with probability q, then X yields $1-q$ and Y yields u_Y. Hence, X is a best response whenever $q \leq 1-u_Y$, and the size of the basin of attraction of X is

$$\mathcal{B} = 1 - u_Y.$$

I refer to $\mathcal{B}$ as the *static basin*. Action X is *risk dominant* when $\mathcal{B} \geq 0.5$, that is, when X is a best response to the belief that the other player is equally likely to choose either action. A larger u_Y shrinks $\mathcal{B}$, so X is optimal for a smaller range of beliefs and coordination on X becomes strategically riskier.

Dynamic basin of attraction. A dynamic analogue must specify which strategies an agent is uncertain about. Comparing unconditional All-X (choose X in every round) with All-Y (choose Y in every round) is problematic in a dynamic game: once an All-X agent observes a Y, continuing to choose X is no longer optimal, yet All-X cannot respond. Following Dal Bó and Fréchette (2011), I instead compare All-Y with *Grim Trigger* (GT), under which an agent chooses X as long as every previous round has been all-X and switches permanently to Y after the first round that is not. GT keeps coordination on X possible while responding optimally to a coordination failure. GT is an SPE, and in this game it

induces the same play as Tit-for-Tat and state-dependent variants of both rules, both on the equilibrium path and after any unilateral deviation.[7]

Suppose that agent i believes that each other member, independently, is a GT type with probability p and an All-Y type with probability $1-p$. Following Boczoń et al. (2025), this measures strategic uncertainty *per opponent*. Let $m \equiv \max\{\bar{K}-k, 1\}$ denote the number of consecutive all-X rounds the group still needs to become protected from state k, truncated at one because a single all-X round secures protection from both $k = \bar{K}-1$ and $k = \bar{K}$. Agent i prefers GT to All-Y at a cooperative history in state k, that is, a history with no coordination failure so far, if and only if $p \geq p_k^\star$,[8] where

$$p_k^\star \equiv \left(\frac{u_Y}{\frac{1-\delta u_Y}{1-\delta} + (1+d)\delta^{m-2}} \right)^{\frac{1}{n-1}}.$$

Definition 2 (Dynamic basin of attraction). The *dynamic basin of attraction* of X at state k (hereafter, the *dynamic basin*) is

$$\mathcal{DB}_k \equiv 1 - p_k^\star,$$

the largest probability that each other member is an All-Y type such that Grim Trigger remains optimal at state k.

A larger dynamic basin means that coordination on X remains optimal under more pessimistic beliefs about the other members. Three comparative statics follow directly. First, $\mathcal{DB}_k$ decreases in u_Y: a more attractive safe action makes Y optimal under a wider range of beliefs. Second, $\mathcal{DB}_k$ decreases in n: GT pays off only if all $n-1$ other members are GT types, which is less likely in a larger group. Third, $\mathcal{DB}_k$ strictly increases in k for $k \leq \bar{K}-1$ and is constant between $k = \bar{K}-1$ and $k = \bar{K}$, because from both states a single all-X round secures protection. Thus, the dynamic basin expands as the group makes progress, but the expansion stops once protection is one round away.

At the initial state, the dynamic basin coincides with the belief cutoff from the monotone mixed MPE in Section 2.2.2: for $\bar{K} = 1, 2, 3$, $p_0^\star$ equals σ_0 in $(m1)$, $(m11)$, and $(m111)$, so $\mathcal{DB}_0 = 1 - \sigma_0$. In both constructions, a failure at $k = 0$ leads to the All-Y continuation value and success leads to the cooperative path, so the indifference conditions are identical.[9]

[7] See Appendix A3.1 for the definitions of these rules, the proof that they induce the same play, and the proof that they are SPEs.

[8] Appendix A3.2 derives this cutoff from the continuation values of the two strategies.

[9] Appendix A3.3 shows the algebra for $\bar{K} = 1, 2, 3$.

The Markov and Grim-Trigger approaches therefore yield the same measure of strategic uncertainty at the initial state.

Figure 2 illustrates the experimental parameterization, $\bar{K} = 2$, $\delta = 0.8$, $d = 1$, $n \in \{2, 4\}$, and $u_Y \in \{0.15, 0.85\}$. With $n = 2$ and $u_Y = 0.15$, the dynamic basin is close to one at all states, so X remains optimal under a wide range of beliefs about the other members. With $n = 4$ and $u_Y = 0.85$ (treatment $4h$ in Section 3), the dynamic basin is below 0.5 at all states, so choosing X requires substantially more optimistic beliefs. This is therefore the most strategically difficult of the four treatment conditions. In every condition, the dynamic basin expands from $k = 0$ to $k = 1$ and is flat from $k = 1$ to $k = 2$.[10]

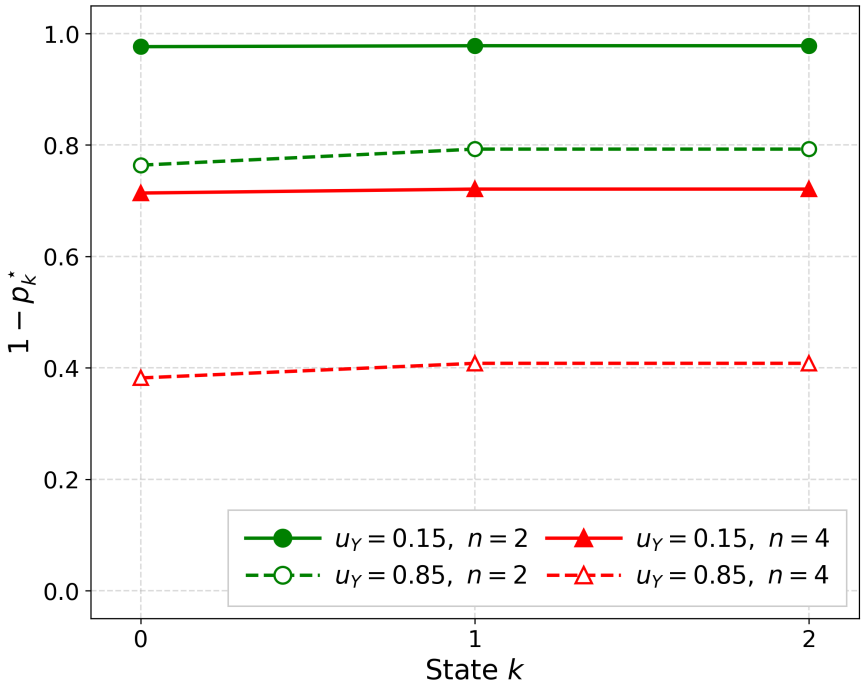


Figure 2: Dynamic Basin of Attraction for Grim Trigger against All-Y

[10] Additional parameter comparisons are reported in Appendix A2.

3 Experimental Design

3.1 Treatments and Parameters

The experiment varies two treatment dimensions: the normalized payoff of the safe action, $u_Y \in \{0.15, 0.85\}$, and group size, $n \in \{2, 4\}$. Group size varies between sessions, whereas each subject experiences both values of u_Y, with the order counterbalanced across sessions. The remaining game parameters are fixed at $\bar{K} = 2$, $d = 1$, and $\delta = 0.8$ across all treatment conditions. Payoffs are implemented in experimental points by multiplying the normalized stage payoffs in Table 1 by 100. Thus, $u_Y = 0.15$ and $u_Y = 0.85$ correspond to 15 and 85 experimental points, respectively. The stage payoffs used in the laboratory are shown in Table 3.

		Group outcome	
		all-X	not all-X
You	X	100	0
	Y		15

(a) When u_Y is low

		Group outcome	
		all-X	not all-X
You	X	100	0
	Y		85

(b) When u_Y is high

Table 3: Stage Payoffs in Experimental Points

The parameter values are chosen to generate meaningful variation in strategic uncertainty while keeping the dynamic structure parsimonious. I set $\bar{K} = 2$ because with $\bar{K} = 1$ there is no intermediate unprotected state: one successful all-X round immediately reaches protection. Moreover, for the monotone mixed-MPE benchmark, increasing the threshold from $\bar{K} = 2$ to $\bar{K} = 3$ mainly extends the sequence of pure-X states following the initial mixing state rather than introducing a qualitatively different pattern. Thus, $\bar{K} = 2$ captures the main transition of interest in a more parsimonious experimental design. I set $\delta = 0.8$, implying an expected match length of five rounds. This is long enough, on average, for a group to reach the threshold, experience a coordination failure and reset, and subsequently reach the threshold again, creating within-match variation useful for distinguishing state from history dependence.

I use $n \in \{2, 4\}$ to generate a substantial change in coordination difficulty while keeping group formation manageable in the laboratory. Although four is not a large group in absolute terms, it doubles the two-player benchmark and creates a stronger theoretical contrast than $n = 3$. In particular, the individual belief threshold depends nonlinearly on group size through the power $1/(n-1)$, so increasing group size from two to four produces a larger change in the per-opponent belief cutoff than increasing it from two to three. I choose

$u_Y \in \{0.15, 0.85\}$ to generate a sharper contrast in the attractiveness of the safe action than the more moderate parameterization $u_Y \in \{0.25, 0.75\}$. Finally, I set $d = 1$ so that the terminal loss equals the normalized payoff from successful coordination in magnitude; together with the 100-point endowment that each subject receives at the beginning of every match, this prevents negative match earnings. Appendix A2 reports the corresponding theoretical comparisons under alternative values of $\bar{K}$, u_Y, δ, and d.

Part 1 is the main game of the experiment and consists of 10 matches. Subjects are randomly rematched before each match. Group size n is fixed throughout a session, while each value of u_Y is used for five consecutive matches, with one payoff condition applying to Matches 1-5 and the other to Matches 6-10. The four Part 1 treatment conditions are denoted $2l$, $2h$, $4l$, and $4h$, where the number denotes group size and l and h denote the low and high safe-payoff conditions, respectively. Accordingly, the four treatment orders are $2l + 2h$, $2h + 2l$, $4l + 4h$, and $4h + 4l$.

Each of the four treatment orders was conducted once under each of the three predrawn termination sequences[11], giving 12 treatment sessions. Table 4 reports the realized number of subjects in each session. The treatment sample contains 164 subjects: 68 in the $n = 2$ sessions and 96 in the $n = 4$ sessions. The planned sample size was 168; two of the $n = 2$ sessions were conducted with 10 rather than 12 subjects due to actual attendance.

After completing the treatment sessions, I conducted two additional baseline sessions as a robustness benchmark. The baseline sessions use the same stage-game payoffs and random-termination procedure but omit the public state counter and the last-round payoff penalty. Both baseline sessions have group size $n = 4$ and use Sequence 2, with the two payoff orders counterbalanced. The baseline sample contains 32 subjects, bringing the total number of subjects across all non-pilot sessions to 196.

3.2 Implementation

The experiment was approved by the Purdue University IRB (STUDY2026-00000142) and conducted at the Vernon Smith Experimental Economics Laboratory (VSEEL) at Purdue University in August and September 2026. The experiment was implemented in oTree 6.0.14 (Chen et al., 2016). Pilot sessions conducted in April 2026 were used to test the software, instructions, and parameterization; pilot data are not included in the analysis.

Subjects were Purdue University undergraduate students from a variety of disciplines and were recruited by email using ORSEE (Greiner, 2015). Each subject participated in only one session and provided electronic informed consent before the experiment began. Given

[11] Appendix C2 lists the three predrawn termination sequences under $\delta = 0.8$.

Table 4: Treatment Orders and Experimental Sessions

Treatment order	Safe payoff u_Y (points)		Sequence			# Sessions	# Subjects
	Matches 1-5	Matches 6-10	1	2	3		
Treatments							
$2l + 2h$	15	85	12	12	10	3	34
$2h + 2l$	85	15	10	12	12	3	34
$4l + 4h$	15	85	16	16	16	3	48
$4h + 4l$	85	15	16	16	16	3	48
Treatment total			54	56	54	12	164
Baseline							
$4l + 4h$	15	85	-	16	-	1	16
$4h + 4l$	85	15	-	16	-	1	16
Total			54	88	54	14	196

Notes: Entries under Sequences 1-3 report the number of subjects in each session. Each treatment was conducted once under each of the three sequences. The baseline sessions use the same stage-game payoffs but omit the public counter and the last-round penalty. For the treatments, $\bar{K} = 2$, $d = 1$, and $\delta = 0.8$ are fixed across treatment conditions.

the importance of common information in coordination games (Chen et al., 2021), subjects were informed on the Overview page that all participants in a session received the same instructions and that deception would not be used.

Each session consisted of three parts. Part 1 contained the 10 matches of the dynamic stag-hunt game described above. Before beginning Part 1, subjects received the instructions and completed a paid six-question comprehension quiz designed to reinforce and assess their understanding of the game. The payoff table for Matches 1-5 differed from that for Matches 6-10 only in the safe-option payoff u_Y. Subjects were informed at the outset that Part 1 contained 10 matches, but the change in the safe action payoff was revealed only after they completed Match 5.

The following procedure applies to the 12 treatment sessions. The two baseline sessions followed the same procedures, but without the public state counter and the last-round payoff penalty. The public state was displayed to subjects as a public counter. Let k_t denote the updated state at the end of round t, with $k_0 = 0$. At the beginning of round t, subject i observes the group-level inherited state k_{t-1}, displayed on the interface as the current public counter. Subject i then uses a slider to assign a probability to the action X, denoted by $p_i(X) \in \{0, \ldots, 100\}$, individually and simultaneously. To identify whether subjects use mixed strategies at each state, I use an elicitation method similar to Romero and Rosokha (2023): Alongside the slider, a 10×10 grid containing 100 numbered cells visualizes the

chosen probabilities: $p_i(X)$ cells are displayed in yellow for X, and the remaining $100-p_i(X)$ cells are displayed in blue for Y. After subject i submits the decision $p_i(X)$, the computer draws an integer uniformly from $\{1,\ldots,100\}$. If the draw falls in a yellow cell, action X is implemented for the subject in that round; otherwise, action Y is implemented. The realized action is shown to the participant. After all group members have submitted their decisions and actions have been realized, subjects observe the group outcome (a list of each member's action), their own round payoff, and the updated state k_t at the end of a round. An all-X outcome increases the state by one until the threshold $\bar{K}=2$ is reached, whereas a non-all-X outcome resets the state to zero. Once the threshold has been reached, another all-X outcome maintains the group at the protected state.

Each match used random termination with continuation probability $\delta=0.8$. The continuation draws were generated in advance as independent draws from the discrete uniform distribution on $\{1,\ldots,100\}$. At the end of each round, the corresponding draw was displayed to subjects through a spinner with an 80% white continuation region and a 20% gray stopping region. A draw at or below 80 continued the match, while a draw above 80 ended it. Subjects knew the continuation probability but did not know the predrawn sequence or, when making their decisions, which round would ultimately be the last. Appendix C2 reports the three predrawn continuation sequences.

In the treatment sessions, if the state at the end of the last round satisfied $k_T<\bar{K}$, every group member's payoff for round T was overridden to -100 experimental points. If $k_T=\bar{K}$, the payoff for the last round was determined by the stage-payoff table as usual. This last-round payoff rule was absent in the baseline sessions.

Part 2 elicits subjects' risk preferences using a multiple-price-list task adapted from Cason et al. (2025). Part 3 measures subjects' social preferences using the Social Value Orientation (SVO) task introduced by Murphy et al. (2011). Subjects receive incentive payments for both Parts 2 and 3. After completing Part 3, subjects answer a short survey collecting demographic information and brief descriptions of the strategies they used in Part 1.

Total earnings from the comprehension quiz and all three parts were displayed only on the final page of the experiment and were paid in cash. The average payment was \$17.15 per subject, with an interquartile range of [\$14.96, \$19.86]. Across the 14 non-pilot sessions, 196 subjects participated: 164 in the 12 main treatment sessions and 32 in the two baseline sessions. Due to actual attendance, the treatment sessions with $n=2$ included 10-12 subjects, while all $n=4$ sessions included 16 subjects. Sessions lasted approximately 45 minutes, including instructions and payment distribution. See Appendix C1 for the experimental instructions, comprehension quiz, game interfaces, and survey.

3.3 Hypotheses

For the experimental parameters $\bar{K} = 2$, $\delta = 0.8$, and $d = 1$, the theoretical framework generates three comparative predictions that guide the experiment. The dynamic basin expands as the public state k increases, is larger when the safe-option payoff u_Y is lower, and is larger in smaller groups. At the same time, unanimous X remains socially efficient in every state and treatment. Thus, the experimental variation changes the strategic difficulty of coordination while holding the efficient action fixed.

Based on the equilibrium analysis and the dynamic basin of attraction derived in Section 2, I formulate three main hypotheses corresponding to those specified in the preregistration. The first concerns how behavior varies with the current public state, the second distinguishes state-dependent from history-dependent strategies, and the third concerns comparative statics in group size and the payoff from the safe action. Note that throughout the empirical analysis, k denotes the state observed when the subject makes the current-round decision, corresponding to k_{t-1} in the time-indexed game. The study was preregistered on AsPredicted (#308239) prior to data collection.

Hypothesis 1 (Monotonicity in state). The probability assigned to X is weakly increasing in the current public state k, and the increase from $k = 0$ to $k = 1$ is greater than the increase from $k = 1$ to $k = 2$.

Both the state-based MPE analysis and the history-based Grim-Trigger benchmark imply that coordination on X becomes more attractive as the group moves closer to the threshold. In the state-based framework, a higher k increases the continuation value associated with successful coordination. In the dynamic-basin framework, a higher k expands the range of beliefs under which continued coordination on X is preferred. Thus, although the two frameworks differ in whether behavior depends only on the current state or also on the history leading to that state, both predict that the probability assigned to X should weakly increase with k.

The stronger prediction that the increase from $k = 0$ to $k = 1$ exceeds the increase from $k = 1$ to $k = 2$ follows from the shape of the dynamic basin: with $\bar{K} = 2$, the basin expands from $k = 0$ to $k = 1$ but not from $k = 1$ to $k = 2$ (Section 2.3, Figure 2). I therefore examine not only whether $p(X)$ increases across states, but also whether

$$p(X \mid k = 1) - p(X \mid k = 0) > p(X \mid k = 2) - p(X \mid k = 1).$$

Hypothesis 2 (State- versus history-dependent strategies). Holding the current public state k fixed, the probability assigned to X does not systematically depend on the history leading to that state. In particular, behavior at $k = 0$ following a reset should be similar to behavior

at the initial $k = 0$ state of a match.

The current public state k is a sufficient statistic for the objective continuation problem. Conditional on the same value of k, the continuation probability, payoff structure, and distance from the threshold are the same regardless of the history through which that state was reached. In particular, after a coordination failure resets the state to $k = 0$, the objective incentives to begin coordinating toward the threshold are the same as at the initial $k = 0$ state of a match.

Moreover, unlike in a repeated prisoner's dilemma, coordination on X does not require a history-dependent punishment to deter a profitable unilateral deviation: when all other group members coordinate on X, choosing X is itself a best response. The current state is also publicly displayed at every decision, making it a salient basis for behavior. These features make state-dependent strategies a natural behavioral benchmark.

Previous coordination experiments show that past play and historical precedents can shape subsequent equilibrium selection (Van Huyck et al., 1991; Rankin et al., 2000; Schmidt et al., 2003). In this paper, the public state allows history dependence to be examined while holding the current payoff-relevant state fixed. History dependence nevertheless remains plausible. A coordination failure may cause subjects to update their beliefs about the willingness of their current group members to coordinate, even though the objective continuation problem has returned to the same state. The distinction is therefore an empirical one rather than a mechanical consequence of the game. Hypothesis 2 tests whether behavior is summarized by the current state or retains systematic dependence on the path through which that state was reached.

Hypothesis 3a (Comparative statics in group size). For a given payoff from the safe action Y, the probability assigned to X is higher in smaller groups.

Successful coordination requires confidence that all other $n - 1$ group members will coordinate on X. Under the dynamic-basin framework, this becomes more difficult as group size increases because each subject must hold sufficiently optimistic beliefs about a larger number of opponents. Equivalently, the per-opponent dynamic basin shrinks with n. This predicted group-size effect is also related to experimental evidence of greater coordination difficulty in larger-group environments (Van Huyck et al., 1990; Weber, 2006). The theoretical comparative statics hold state by state. Experimentally, Hypothesis 3a implies two pairwise comparisons:

$$p(X \mid 2l) > p(X \mid 4l) \qquad \text{and} \qquad p(X \mid 2h) > p(X \mid 4h).$$

Hypothesis 3b (Comparative statics in safe-option payoff). For a given group size, the probability assigned to X is higher when the payoff from choosing Y is lower.

In the dynamic-basin analysis, a higher payoff from the safe action Y makes persistent play of Y more attractive and reduces the range of beliefs under which coordination on X is preferred. Hence, the dynamic basin is larger when u_Y is lower. Experimentally, Hypothesis 3b implies two pairwise comparisons:

$$p(X \mid 2l) > p(X \mid 2h) \qquad \text{and} \qquad p(X \mid 4l) > p(X \mid 4h).$$

Together, Hypotheses 3a and 3b provide four pairwise comparisons that vary one experimentally manipulated parameter at a time.

As an additional implication of Hypothesis 1, I examine whether coordination stabilizes once the group reaches the threshold $k = \bar{K}$. At the threshold, successful coordination protects the group from the terminal overwrite loss, so deviations to Y should become relatively rare. I treat this protection-stability comparison as a secondary analysis rather than as a separate main hypothesis.

The primary outcome is subject i's stated probability of choosing X. For the empirical analysis, I divide the elicited 0-100 probability by 100 and denote the resulting measure by $p_{isgmr}(X) \in [0, 1]$, where s indexes sessions, g groups, m matches, and r rounds. The realized individual action and group outcome are treated as secondary outcomes because they additionally reflect the random draws used to implement subjects' stated probabilities. The empirical analysis follows the order of the hypotheses, examining state dependence, state versus history dependence, and the comparative statics in group size and the payoff from the safe action. The corresponding regression specifications are introduced together with the results in Section 4. For round-level regressions, inference is based on wild cluster bootstrap-t tests clustered at the session level with 5,000 bootstrap replications. The group-size comparisons use session-level randomization inference because group size is assigned at the session level. Tests of the directional hypotheses are one-sided at the 5% significance level, while the test of state versus history dependence is two-sided.

4 Results

Before turning to the main results, Table 5 summarizes subject characteristics and elicited preferences by group-size treatment. Participants were randomly assigned to sessions with $n = 2$ or $n = 4$. The table reports treatment-group means and t-statistics for differences in means or proportions as descriptive measures of realized balance across the two conditions. The reported characteristics are broadly similar across group-size treatments. Attendance at an English-speaking high school is included as a proxy for prior exposure to an English-speaking environment and, more broadly, cultural background. Social preference is coded as a binary indicator equal to 1 for Prosocial and 0 for Individualistic. Only one of the 164 treatment participants is classified as Altruistic, and that subject is included in the Prosocial category.

Table 5: Balance of Subject Characteristics and Elicited Preferences

Subject characteristic	$n = 2$		$n = 4$		t-statistic
	Mean	(SE)	Mean	(SE)	
Gender (1 = Male)	0.53	(0.06)	0.49	(0.05)	0.50
Major (1 = Engineering/Science)	0.57	(0.06)	0.57	(0.05)	0.01
High school language (1 = English)	0.85	(0.04)	0.88	(0.03)	-0.41
Risk-averse (1 = Yes)	0.44	(0.06)	0.44	(0.05)	0.05
Social Preference (1 = Prosocial)	0.50	(0.06)	0.61	(0.05)	-1.46
Sessions	6		6		
Subjects	68		96		

Notes: Standard errors are shown in parentheses. All characteristics are binary variables. The t-statistics for differences in means or proportions are reported as $n = 2$ minus $n = 4$.

4.1 Monotonicity in state

Hypothesis 1 concerns the monotonicity of the probability assigned to X in state k. To formally test Hypothesis 1, I estimate

$$p_{isgmr}(X) = \alpha_i + \mu_{sgm} + \lambda_r + \beta_1 1\{k = 1\} + \beta_2 1\{k = 2\} + \varepsilon_{isgmr}, \tag{1}$$

where α_i denotes subject fixed effects, μ_{sgm} group-by-match fixed effects, and λ_r round-within-match fixed effects. The omitted category is state $k = 0$. Thus, β_1 measures the difference between $k = 1$ and $k = 0$, β_2 measures the difference between $k = 2$ and $k = 0$, while $\beta_2 - \beta_1$ measures the difference between $k = 2$ and $k = 1$. Hypothesis 1 predicts that both β_1 and β_2 are weakly positive and that the increase from $k = 0$ to $k = 1$ exceeds that

from $k = 1$ to $k = 2$, corresponding to $2\beta_1 - \beta_2 > 0$. I also estimate a specification replacing the state indicators with the linear state variable k. Because the public state k is generated by previous play rather than experimentally assigned, these estimates are interpreted as conditional associations rather than causal effects of k.

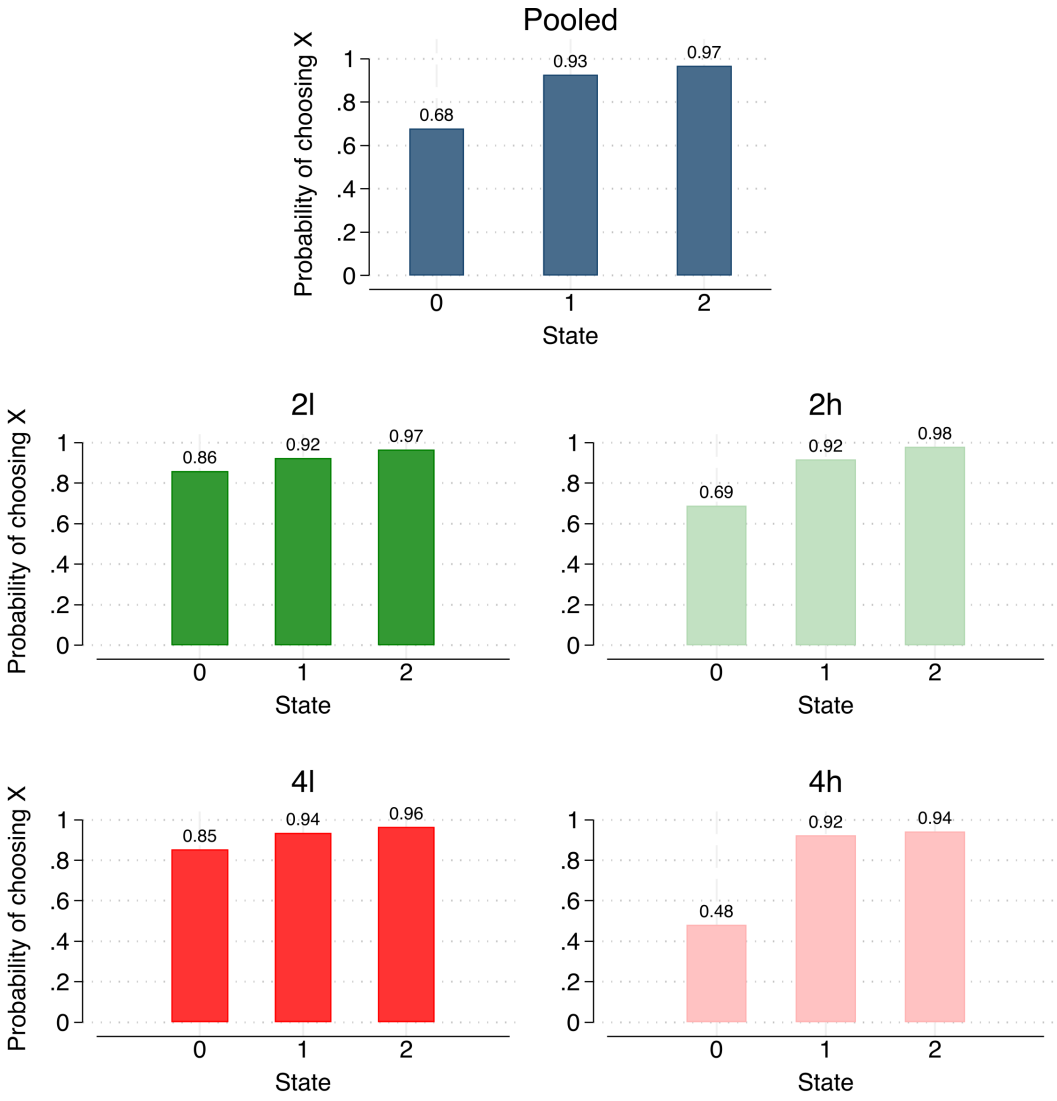


Figure 3: Probability Assigned to X by State

Figure 3 shows the mean $p(X)$ by state in the pooled sample and across the four treatments.[12] In every treatment, $p(X)$ is lowest at $k = 0$ and close to one at $k = 1$ and $k = 2$. Table 6 relates these differences at $k = 0$ to the theoretical belief cutoffs. Under the experimental parameterization, the $(m11)$ mixing probability σ_0 coincides with the Grim-Trigger

[12]Figure B2.1 in Appendix B2 shows the corresponding pattern for realized all-X coordination.

cutoff $p_0^\star$, providing a common state-0 benchmark for the two theoretical approaches.[13] The treatment ordering of observed $p(X)$ at $k = 0$ largely follows the ordering of the dynamic basin: X is chosen most often in $2l$ and least often in $4h$. The main departure is that $4l$ exhibits more X than $2h$, although its basin is slightly smaller (0.71 versus 0.76). The comparisons in Figure 3 and Table 6 are descriptive. Because group-matches with higher underlying propensities to choose X are more likely to reach higher states, the fixed-effects regressions below isolate within-group-match state variation and can yield smaller, or even differently signed, state gradients.

Table 6: Theoretical Cutoffs and Observed Behavior at $k = 0$

Treatment	Belief cutoff $\sigma_0 = p_0^\star$	Dynamic basin $\mathcal{DB}_0 = 1 - \sigma_0$	Observed mean $p(X)$ at $k = 0$
$2l$	0.023	0.977	0.86
$2h$	0.236	0.764	0.69
$4l$	0.286	0.714	0.85
$4h$	0.618	0.382	0.48

Notes: Theoretical values use $\bar{K} = 2$, $\delta = 0.8$, and $d = 1$. The cutoff $\sigma_0 = p_0^\star$ is the minimum probability assigned to X by each other group member at $k = 0$ for X to be a best response. Observed values are mean elicited probabilities assigned to X at $k = 0$, pooling initial and post-reset observations as in Figure 3.

Table 7 reports the tests of Hypothesis 1. The probability assigned to X increases by 0.064 from $k = 0$ to $k = 1$ ($p = 0.002$), and by 0.033 from $k = 1$ to $k = 2$ ($p = 0.073$). The overall difference between $k = 0$ and $k = 2$ is 0.097 ($p = 0.004$). The estimated difference between the two adjacent increments is 0.030 in the predicted direction, but falls just short of statistical significance at the 5% level ($p = 0.056$). Consistent with the overall monotonic pattern, the linear-state specification also yields a positive and statistically significant coefficient of 0.049 ($p = 0.006$).

Table 8 reports the same specifications separately by treatment. The state relationship varies substantially across treatment conditions. The increase from $k = 0$ to $k = 1$ is positive and statistically significant in 2h, 4l, and 4h, but is close to zero in 2l. In contrast, the increase from $k = 1$ to $k = 2$ is not statistically significant in any treatment. The difference between the two increments is positive in all four treatments, although it is statistically significant at the 5% level only in 2h and at the 10% level in 2l. Consistent with this heterogeneity, the linear-state coefficient is positive and statistically significant in 2h and 4h, but not in 2l or 4l. Overall, the monotonic state relationship is strongest in the high-u_Y treatments.

The observed state pattern is also qualitatively consistent with the ($m11$) mixed-MPE

[13] See Section 2.2.2 for the ($m11$) mixed-MPE benchmark and Section 2.3 for the Grim-Trigger cutoff.

Table 7: Monotonicity in the Public State

Contrast	Estimate	Clustered SE	One-sided p	95% CI
Panel A: Categorical state specification				
$k = 1$ minus $k = 0$	0.064	0.014	0.002	[0.032, 0.095]
$k = 2$ minus $k = 0$	0.097	0.033	0.004	[0.027, 0.170]
$k = 2$ minus $k = 1$	0.033	0.022	0.073	[-0.013, 0.084]
$(k = 1 - k = 0) - (k = 2 - k = 1)$	0.030	0.018	0.056	[-0.010, 0.066]
Panel B: Linear state specification				
One-unit increase in k	0.049	0.016	0.006	[0.012, 0.085]
Observations		7,812		
Subjects		164		
Sessions		12		

Notes: All specifications include subject, group-by-match, and round-within-match fixed effects. Analytic weights give each of the 12 sessions equal total weight. Standard errors and wild-cluster-bootstrap-t tests are clustered by session; bootstrap inference uses 5,000 replications. Reported p-values are one-sided in the direction predicted by Hypothesis 1; confidence intervals are two-sided 95% wild-bootstrap intervals. Alternative specifications replacing subject fixed effects with observed participant controls are reported in Appendix B1, Table B1.1.

Table 8: Monotonicity in the Public State by Treatment

	$2l$	$2h$	$4l$	$4h$
	(1)	(2)	(3)	(4)
Panel A: Categorical state specification				
$k = 1$ minus $k = 0$	-0.005	0.066**	0.024**	0.099***
	(0.036)	(0.014)	(0.014)	(0.033)
$k = 2$ minus $k = 0$	-0.055	0.065**	0.044	0.168**
	(0.050)	(0.032)	(0.037)	(0.076)
$k = 2$ minus $k = 1$	-0.049	-0.001	0.019	0.069
	(0.017)	(0.022)	(0.028)	(0.057)
Difference between increments	0.044*	0.067**	0.005	0.030
	(0.026)	(0.018)	(0.025)	(0.055)
Panel B: Linear state specification				
One-unit increase in k	-0.026	0.034**	0.022	0.084**
	(0.025)	(0.016)	(0.018)	(0.037)
Observations	1,620	1,616	2,288	2,288
Subjects	68	68	96	96
Sessions	6	6	6	6

Notes: All specifications include subject, group-by-match, and round-within-match fixed effects. Analytic weights give each of the six sessions represented in a treatment column equal total weight. Session-clustered standard errors are in parentheses. Stars use one-sided wild-cluster-bootstrap-t tests with 5,000 replications in the direction predicted by Hypothesis 1: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Columns denote group size and the low or high safe-option payoff.

benchmark characterized in Section 2. For the experimental threshold $\bar{K} = 2$, this equilibrium prescribes interior mixing at $k = 0$ and pure X at $k = 1$ and $k = 2$. Consistent with this qualitative pattern, the probability assigned to X is substantially lower at $k = 0$ and rises sharply once the state reaches $k = 1$, with relatively little additional increase from $k = 1$ to $k = 2$. The observed probabilities at the higher states are nevertheless below one, so the aggregate state pattern should be interpreted as resembling, rather than exactly implementing, the $(m11)$ equilibrium.

As a secondary analysis, Table B1.1 in Appendix B1 examines heterogeneity by elicited risk and social preferences. Risk aversion is not significantly associated with $p(X)$, while prosociality is positively associated with $p(X)$, although the latter relationship is only weakly statistically significant and is sensitive to the inclusion of additional participant controls.

Result 1: Consistent with Hypothesis 1, the probability assigned to X increases with the public state k in the pooled sample. The increase from $k = 0$ to $k = 1$ is larger than that from $k = 1$ to $k = 2$ in magnitude, but the difference is not statistically significant at the 5% level. The strength of the state relationship varies across treatments.

4.2 State-dependent and history-dependent strategies

Hypothesis 2 distinguishes state-dependent from history-dependent behavior. Under a state-dependent strategy, subjects condition their choices only on the current public state k, so two observations with the same value of k should generate similar behavior regardless of the path by which that state was reached. By contrast, a history-dependent strategy allows behavior to depend on previous play even after conditioning on the current state.

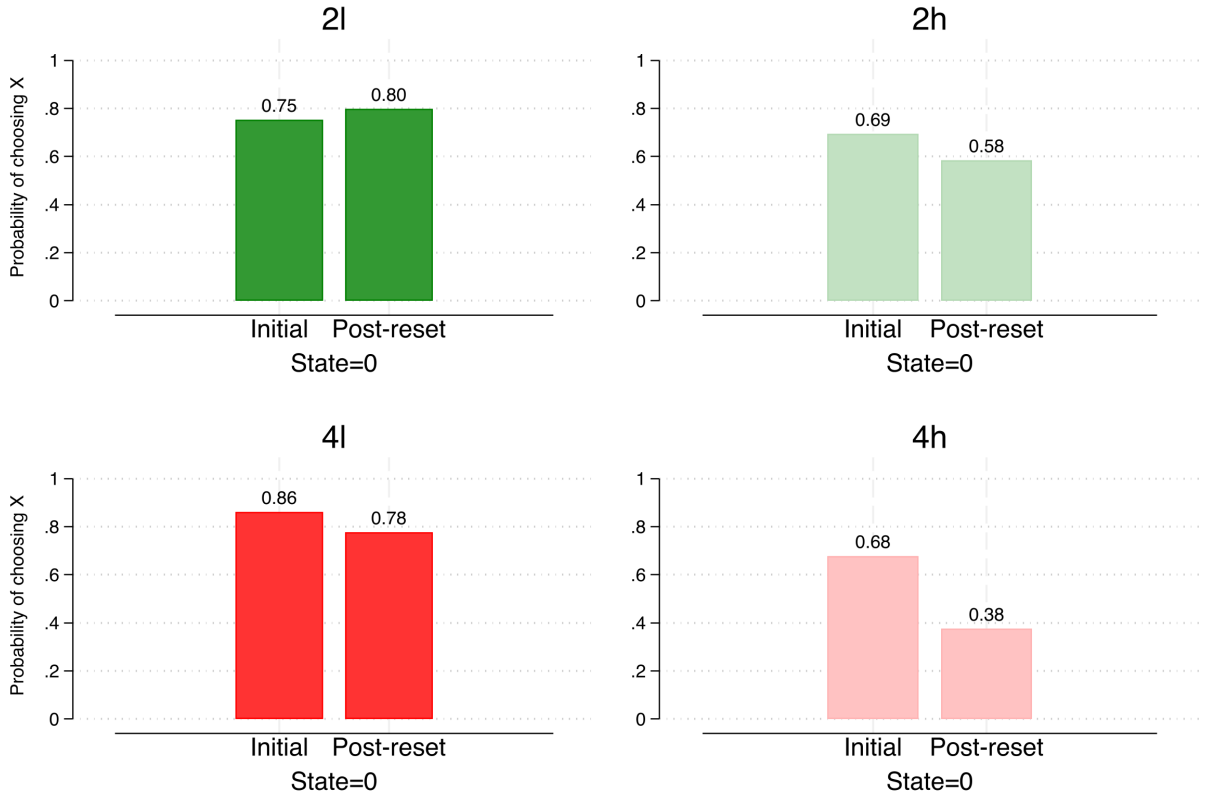
To test Hypothesis 2, I focus on observations with $k = 0$ because this state can arise in two distinct ways: at the beginning of a match, i.e., Round 1, when no previous history has yet accumulated, and after a reset following a failure to coordinate on all-X. Comparing these two cases therefore provides a direct test of whether behavior depends on history beyond the information contained in the current state. I estimate

$$p_{isgmr}(X) = \alpha_i + \mu_{sgm} + \theta Reset0_{sgmr} + \varepsilon_{isgmr}, \qquad k = 0, \tag{2}$$

where $Reset0_{sgmr} = 1$ for a post-reset $k = 0$ observation and $Reset0_{sgmr} = 0$ for the initial $k = 0$ observation. Under state-dependent behavior, the same state should induce the same behavior regardless of the preceding history, implying $\theta = 0$. Hypothesis 2 is therefore evaluated using a two-sided test of $\theta = 0$. Note that round-within-match fixed effects are not included because the initial $k = 0$ observation necessarily occurs in Round 1, whereas

post-reset observations occur only in later rounds.

Figure 4 provides a descriptive comparison of initial and post-reset behavior among group-by-match cells containing both the initial Round-1 $k = 0$ observation and at least one post-reset $k = 0$ observation. This restriction is necessary because some matches end after only one round and others never return to $k = 0$ after Round 1, and it ensures that the initial and post-reset means are calculated over the same set of group-by-match cells. The descriptive pattern is heterogeneous across treatments. In $2l$, mean $p(X)$ is slightly higher after a reset, whereas it is lower after a reset in $2h$, $4l$, and especially $4h$. The increase in $2l$ and the large decline in $4h$ are consistent in direction with the corresponding fixed-effect estimates reported in Table 9.



Notes: The sample is restricted to group-by-match cells containing both the initial Round-1 $k = 0$ observation and at least one post-reset $k = 0$ observation.

Figure 4: Mean $p(X)$ at Initial and Post-Reset $k = 0$ by Treatment

Unlike Figure 4, which restricts the sample to group-by-match cells containing both initial and post-reset $k = 0$ observations, Table 9 reports the corresponding fixed-effect estimates using the full $k = 0$ estimation sample. In the pooled specification, the post-reset coefficient is -0.052 and is not statistically significant, so I do not reject equality between

initial and post-reset behavior. The treatment-specific estimates, however, reveal substantial heterogeneity. Post-reset $p(X)$ is 8.9 percentage points higher in the $2l$ treatment ($p < 0.05$) and 22.6 percentage points lower in the $4h$ treatment ($p < 0.01$); the estimates in $2h$ and $4l$ are small and not statistically significant. Thus, the pooled estimate is consistent with history independence, while the treatment-specific estimates indicate significant departures from it in opposite directions. Because each treatment-specific estimate is based on six sessions, these results should be interpreted cautiously.

The treatment-specific estimates line up with the size of the dynamic basin. Ordered from the largest to the smallest basin at $k = 0$ ($2l$, $2h$, $4l$, and $4h$; Table 6), the post-reset coefficients are 0.089, -0.015, -0.025, and -0.226. One interpretation is that a coordination failure leads subjects to revise downward their beliefs about their current partners. When the dynamic basin is small, as in $4h$, choosing X already requires optimistic beliefs, so even a modest downward revision can make Y the better response; when the basin is close to one, as in $2l$, the same revision leaves X optimal. With only four treatment conditions, this ordering is descriptive rather than a formal test.

Table 9: Initial versus Post-Reset Behavior at $k = 0$

	Probability assigned to X				
	Pooled	$2l$	$2h$	$4l$	$4h$
	(1)	(2)	(3)	(4)	(5)
Post-reset $k = 0$	-0.052	0.089**	-0.015	-0.025	-0.226***
	(0.033)	(0.035)	(0.054)	(0.015)	(0.026)
Observations	4,000	554	714	1,000	1,732
Subjects	164	68	68	96	96
Sessions	12	6	6	6	6
Subject fixed effects	Yes	Yes	Yes	Yes	Yes
Group-by-match fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: Initial $k = 0$ is the omitted category. All specifications include subject and group-by-match fixed effects. Analytic weights give each session equal total weight. Session-clustered standard errors are in parentheses. Stars use two-sided wild-cluster-bootstrap-t tests with 5,000 replications: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Result 2: The pooled data do not reject state dependence, but treatment-specific estimates reveal significant history dependence in opposite directions, with post-reset p(X) higher in the $2l$ treatment and substantially lower in the $4h$ treatment. Post-reset behavior does not resemble the permanent switch to Y prescribed by Grim Trigger.

4.3 Comparative statics in u_Y and n

I next examine the comparative statics in group size n and the payoff from the safe action u_Y. The experimental design varies group size between sessions and u_Y within subjects, so the two treatment dimensions are analyzed separately.

Hypothesis 3a predicts a higher probability of choosing X in smaller groups, i.e., when group size $n = 2$. Because group size is assigned at the session level, I conduct the primary comparison at the session level, separately for each value of u_Y. For each subject, I first calculate the mean probability assigned to X under a given payoff condition and then average these subject-level means within each session. Let $\bar{p}_{su}$ denote this session-level mean for payoff condition $u \in \{l, h\}$. I estimate

$$\bar{p}_{su} = \alpha_u + \beta_{n,u} 1\{n_s = 2\} + \varepsilon_{su}, \qquad u \in \{l, h\}, \tag{3}$$

where $\beta_{n,l}$ compares treatments $2l$ and $4l$, while $\beta_{n,h}$ compares treatments $2h$ and $4h$. Hypothesis 3a predicts $\beta_{n,u} > 0$. Inference is based on session-level randomization tests.

Table 10: Effect of Group Size on the Probability Assigned to X

	Low u_Y		High u_Y	
	Unadjusted (1)	Adjusted (2)	Unadjusted (3)	Adjusted (4)
$n = 2$ minus $n = 4$	0.012	0.012	0.240**	0.240**
	(0.036)	(0.023)	(0.086)	(0.085)
High-first	No	Yes	No	Yes
Sequence fixed effects	No	Yes	No	Yes
Sessions	12	12	12	12

Notes: Standard errors are in parentheses. Stars denote one-sided exact session-level permutation p-values, obtained by swapping the $n = 2$ and $n = 4$ labels within each of the six sequence-by-order pairs: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10 reports the corresponding regression results. The outcome is constructed by first averaging $p(X)$ within subject and payoff condition and then averaging across subjects within each session. Each column contains 12 session observations. Columns (2) and (4) additionally control for payoff order (High-first) and predetermined termination-sequence fixed effects, accounting for variation associated with treatment order and termination sequences. The comparisons do not condition on the realized state. Coefficients are reported as $n = 2$ minus $n = 4$, so positive estimates support Hypothesis 3a. When u_Y is low, the estimated difference between $n = 2$ and $n = 4$ is only 1.2 percentage points and is not statistically significant. In contrast, when u_Y is high, participants in groups of two assign, on average,

a 24.0 percentage-point higher probability to X than participants in groups of four. This difference is statistically significant at the 5% level based on the one-sided exact session-level permutation test. The estimates are unchanged after controlling for payoff order and termination-sequence fixed effects.

Result 3a: The group-size comparative static in Hypothesis 3a is supported when the safe payoff is high, but not when it is low. Smaller groups assign substantially higher probabilities to X under high u_Y, whereas there is no detectable group-size difference under low u_Y.

Hypothesis 3b predicts that lowering the payoff from the safe action increases the probability assigned to X. Since each subject experiences both payoff conditions, I exploit within-subject variation and estimate the following specification separately for $n = 2$ and $n = 4$:

$$p_{isgmr}(X) = \alpha_i + \lambda_m + \beta_{Y,n} LowY_{sm} + \varepsilon_{isgmr}, \qquad n \in \{2, 4\}, \tag{4}$$

where $LowY_{sm}$ equals one when $u_Y = 0.15$, α_i denotes subject fixed effects, and λ_m denotes match-number fixed effects. Thus, $\beta_{Y,n}$ measures the within-subject effect of moving from the high to the low safe-option payoff. Hypothesis 3b predicts $\beta_{Y,n} > 0$.

Table 11: Effect of the Safe Option Payoff on the Probability Assigned to X

	$n = 2$			$n = 4$		
	(1)	(2)	(3)	(4)	(5)	(6)
Low u_Y minus High u_Y	0.100***	0.100*	0.151	0.327***	0.327***	0.211
	(0.028)	(0.064)	(0.037)	(0.036)	(0.053)	(0.085)
Subject fixed effects	Yes	Yes		Yes	Yes	
Absolute match-number FE	Yes		Yes	Yes		Yes
Relative match-number FE		Yes			Yes	
First-half sample			Yes			Yes
Observations	3,236	3,236	1,580	4,576	4,576	2,240
Subjects	68	68	68	96	96	96
Sessions	6	6	6	6	6	6
R^2	0.383	0.325	0.067	0.385	0.371	0.086

Notes: Session-clustered standard errors are in parentheses. Stars use one-sided wild-cluster-bootstrap-t tests in columns (1), (2), (4), and (5), and one-sided exact session-level permutation tests in columns (3) and (6). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11 reports the corresponding estimates separately for $n = 2$ and $n = 4$. Positive coefficients indicate a higher probability assigned to X when $u_Y = 0.15$ than when $u_Y = 0.85$. Columns (1), (2), (4), and (5) exploit within-subject variation in u_Y and do not condition

on the realized state k. Columns (1) and (4) include absolute match-number fixed effects, where the absolute match number is the match number in Part 1 (1-10). Columns (2) and (5) instead include relative match-number fixed effects, where the relative match number is the position within each five-match payoff block (1-5), so Matches 6-10 are recoded as 1-5 after the safe-option payoff changes. As a robustness check for potential order or carryover effects, Columns (3) and (6) provide a between-session comparison using only Matches 1-5, before participants have been exposed to the alternative payoff condition.

Columns (1) and (4) report the primary within-subject estimates. Lowering u_Y increases the probability assigned to X by 0.100 in groups of two and by 0.327 in groups of four, with both effects statistically significant at the 1% level. With relative match-number fixed effects in Columns (2) and (5), the point estimates are unchanged, although the estimate for $n = 2$ is statistically significant only at the 10% level, while the estimate for $n = 4$ remains significant at the 1% level. In the first-half sample in Columns (3) and (6), the estimated effects remain positive, at 0.151 for $n = 2$ and 0.211 for $n = 4$, but neither is statistically significant under the exact session-level permutation tests.

Result 3b: Consistent with Hypothesis 3b, reducing the payoff from the safe action significantly increases the probability assigned to X for both group sizes in the primary within-subject specifications. The corresponding first-half estimates are also positive but are not statistically significant.

4.4 Persistence of Coordination

As a secondary implication of Hypothesis 1, I examine whether coordination remains stable once a group reaches the threshold $k = 2$. Because unanimous X is socially efficient at every state, reaching the threshold corresponds to sustaining the efficient coordination path for two consecutive rounds. I therefore examine how quickly groups reach the threshold and whether all-X coordination persists thereafter.

Figure 5 shows how quickly group-matches reach the threshold, $k = 2$. Among group-matches that reach the threshold, coordination typically emerges rapidly: in all four treatments, most reach $k = 2$ in the minimum possible two rounds. Notably, the $4h$ treatment, which has the smallest dynamic basin in Figure 2, also exhibits the highest rate of failure to reach the threshold.

Figure 6 decomposes the groups that do not reach $k = 2$, where some are mechanical because a match that ends after one round cannot reach the threshold. However, in every treatment, the majority of non-attainment occurs in matches lasting at least two rounds. This pattern is particularly pronounced in the $4h$ treatment, where 86% of the group-matches

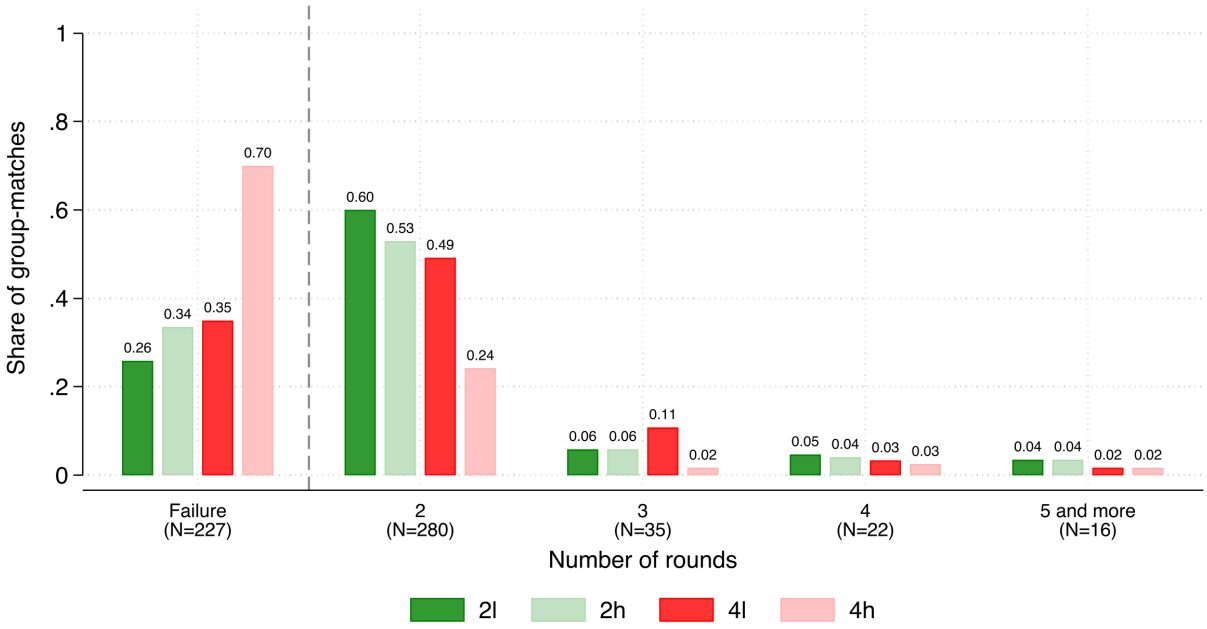


Figure 5: Number of Rounds to First Reach the Threshold

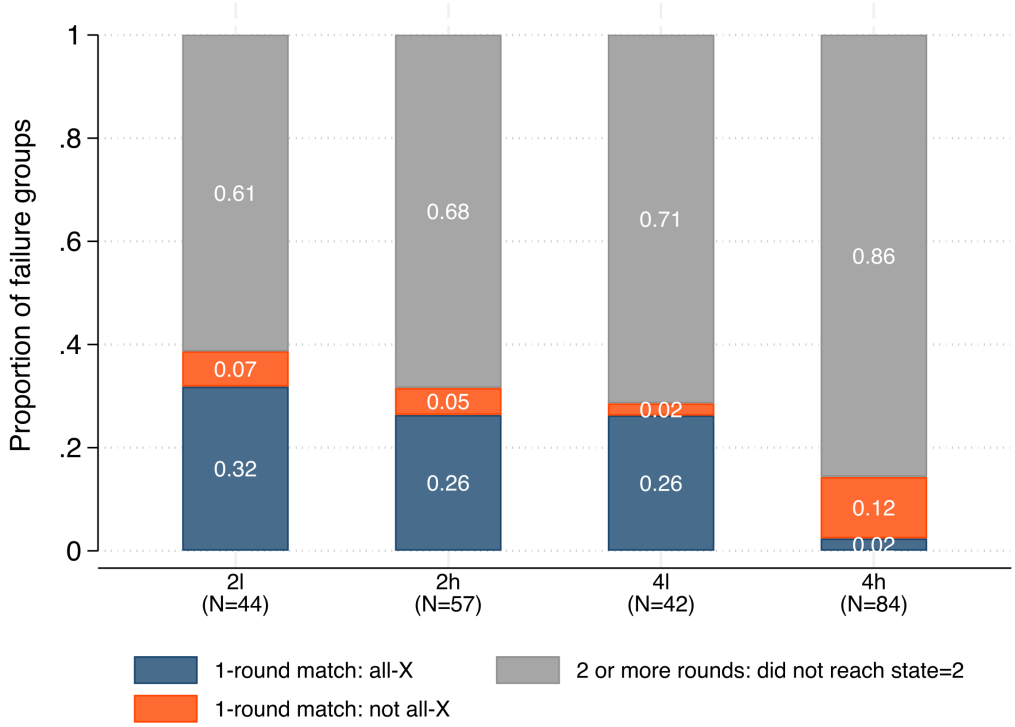


Figure 6: Composition of Groups That Do Not Reach the Threshold

that do not reach $k = 2$ last for at least two rounds. Thus, the relatively high rate of non-attainment in this treatment cannot be explained solely by early random termination.[14]

Figure 7 examines behavior after a group reaches the threshold. Most threshold-reaching group-matches maintain all-X coordination for the remainder of the match: 71%, 74%, 61%, and 56% in the 2l, 2h, 4l, and 4h treatments, respectively. An additional 11-17% reach $k = 2$ only in the final round, so subsequent persistence cannot be observed. Conditional on the match continuing for at least one additional round after the threshold is first reached, approximately 67-84% maintain coordination through the end of the match. The $4h$ treatment also has the lowest rate of maintaining all-X coordination through the remainder of the match.[15]

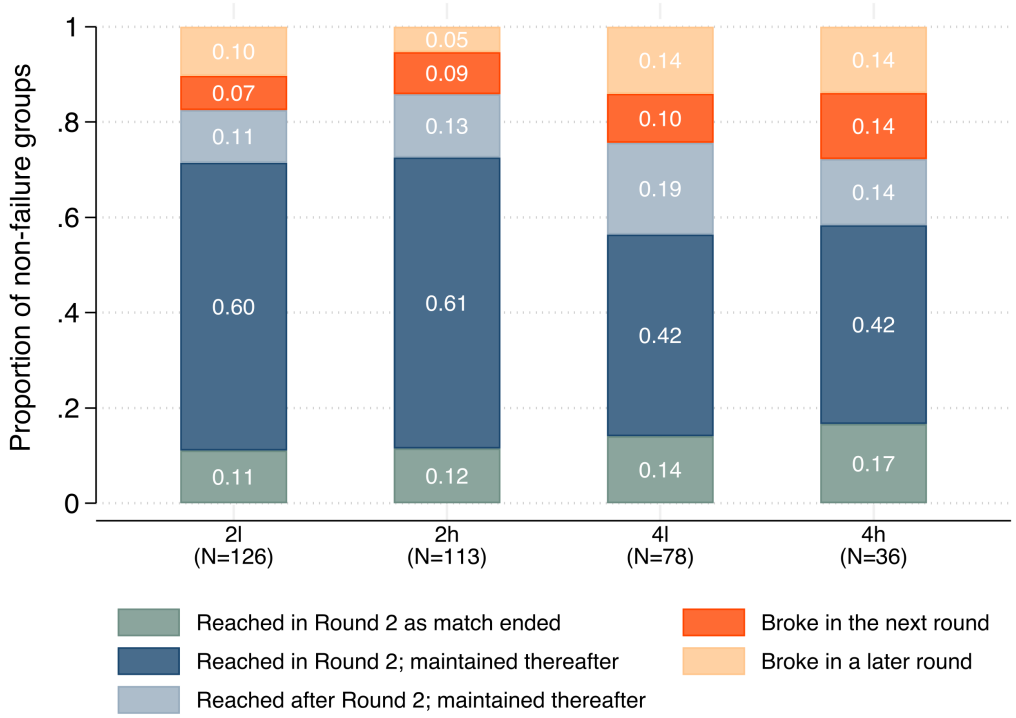


Figure 7: Outcomes after Reaching the Threshold

Result 4: Consistent with the secondary implication of Hypothesis 1, coordination is highly persistent once groups reach the threshold. Successful groups typically reach $k = 2$ in the minimum possible two rounds, and most observable post-threshold paths maintain

[14]The corresponding decomposition by experimental half is reported in Appendix B1, Figure B1.1.
[15]The corresponding decomposition by experimental half is reported in Appendix B1, Figure B1.2.

coordination through the end of the match. Failure to reach the threshold is most frequent in the 4h treatment.

4.5 Mixed Behavior

A feature of the experimental design is that subjects report a probability of choosing X, rather than making a binary choice between actions X and Y. This elicitation can distinguish pure decisions, $p(X) = 0$ or $p(X) = 1$, from mixed behavior, $0 < p(X) < 1$. Such behavior would not be directly observable from binary choices alone, since an observed action does not reveal whether it was chosen with certainty or generated from an interior choice probability.

Figure 8 shows the shares of pure Y, mixed, and pure X decisions at each state. Mixing is most common at $k = 0$, where about 30% of decisions involve interior probabilities. As the state increases, the share of mixed decisions falls to 12% at $k = 2$, while pure X rises from 52% to 87% and pure Y becomes rare. Thus, higher states are associated with a shift from mixed choices toward pure X.

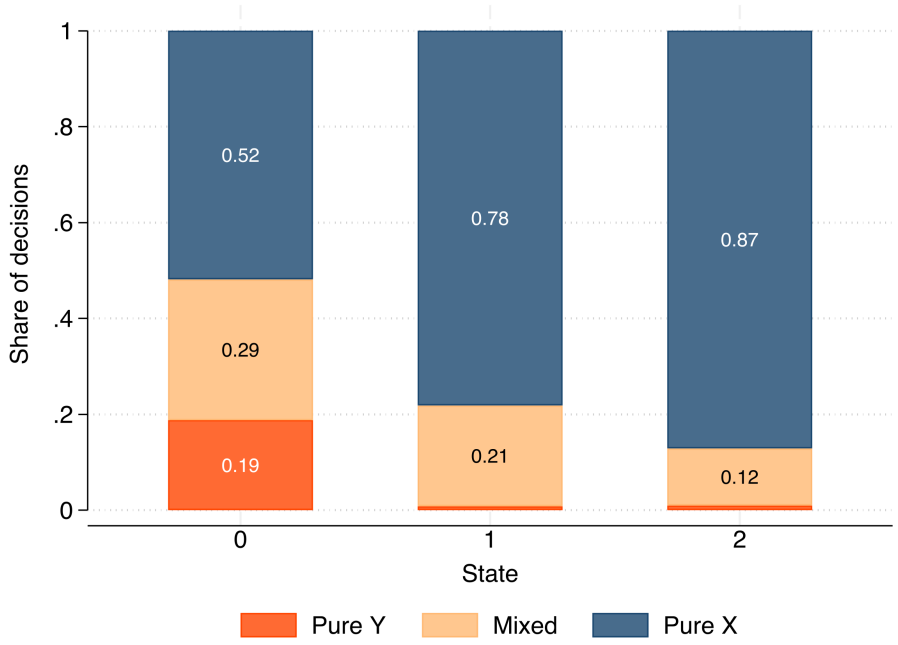


Figure 8: Pure and Mixed Decisions by State

The state comparison does not reveal whether mixed behavior also changes as subjects gain experience with the game. I therefore next examine how the prevalence of pure and mixed choices evolves across matches, both in the pooled sample and separately across treatments. Figure 9 shows how pure and mixed choice behavior evolves over the experiment.

Mixed choices are common initially, accounting for about one-third of decisions in Match 1, but their share declines over subsequent matches. Over the same period, the share of pure X choices increases substantially, whereas the share of pure Y choices remains relatively small. Thus, the decline in mixing is primarily associated with convergence toward pure X rather than pure Y.

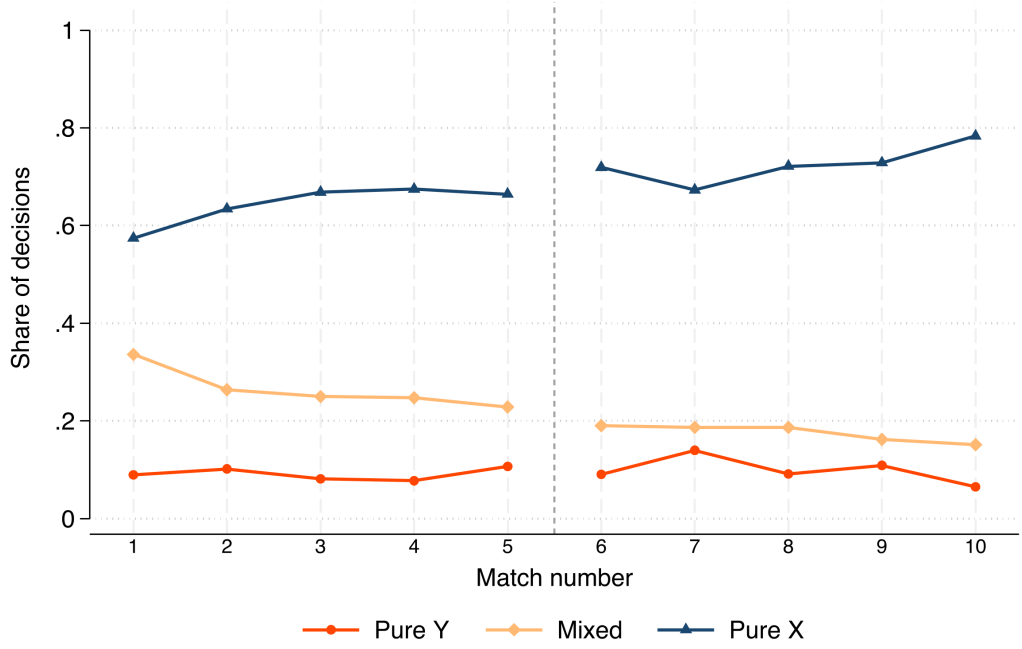


Figure 9: Proportion of Decisions by Match

Figure 10 separates the evolution of mixed behavior by treatment. The decline in mixing is not concentrated in a single treatment: mixed choices generally become less frequent over matches. However, both their initial prevalence and the magnitude of the decline vary across treatments. The treatment-specific patterns therefore suggest a common experience-related decline in mixing with substantial heterogeneity across experimental conditions.

Table 12 examines learning in pure and mixed behavior over repeated play. I use match number within each five-match treatment block as a measure of learning; the variable runs from 1 to 5 and resets after u_Y changes following Match 5. Columns (1)-(3) use an indicator for a mixed decision as the dependent variable, while columns (4)-(6) use an indicator for a pure X decision. Across specifications, mixed behavior declines with learning. In the preferred specification in column (3), an additional match within the block is associated with a 0.9 percentage-point reduction in the probability of mixing ($p < 0.05$). Mixing also declines substantially with the public state: relative to $k = 0$, the probability of mixing is 5.4 percentage points lower at $k = 1$ and 8.1 percentage points lower at $k = 2$.

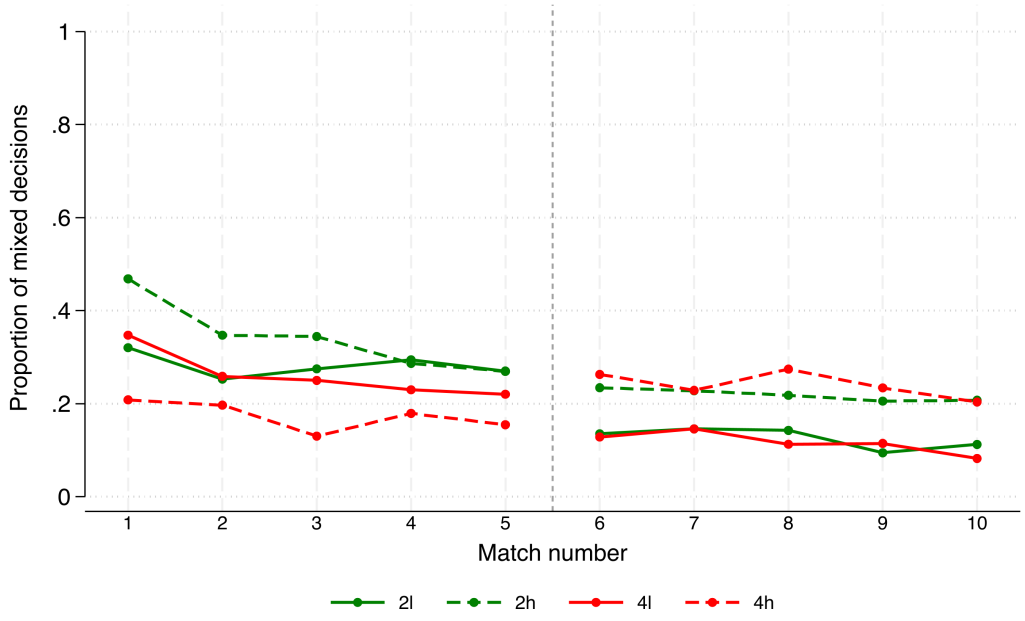


Figure 10: Proportion of Mixed Decisions by Match and Treatment

Table 12: Learning and Mixed Decisions

	Mixed decision			Pure X decision		
	(1)	(2)	(3)	(4)	(5)	(6)
Match within block	-0.007*	-0.006*	-0.009**	0.005	0.004	0.005
	(0.004)	(0.003)	(0.004)	(0.009)	(0.005)	(0.005)
State = 1		-0.067***	-0.054***		0.306***	0.258***
		(0.027)	(0.019)		(0.034)	(0.025)
State = 2		-0.106***	-0.081***		0.426***	0.360***
		(0.034)	(0.019)		(0.049)	(0.040)
Low u_Y			-0.031			0.145***
			(0.025)			(0.033)
Second half			-0.083***			0.056
			(0.026)			(0.035)
Subject FE	Yes	Yes	Yes	Yes	Yes	Yes
Round-within-match FE	Yes	Yes	Yes	Yes	Yes	Yes
Equal-session weights	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,812	7,812	7,812	7,812	7,812	7,812
R^2	0.620	0.628	0.638	0.442	0.542	0.567

Notes: Session-clustered standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Analytic weights give every session equal total weight. Stars use two-sided wild-cluster-bootstrap-t tests with 5,000 replications. All specifications include subject and round-within-match fixed effects. State = 0, High u_Y, and the first half (Matches 1-5) are the omitted categories.

The corresponding estimates for pure X behavior show a different pattern. Conditional on the public state and treatment controls, the within-block learning trend is small and statistically insignificant. In contrast, the state effects are large: relative to $k = 0$, the probability of choosing pure X is 25.8 percentage points higher at $k = 1$ and 36.0 percentage points higher at $k = 2$. Thus, learning is associated with a declining propensity to mix, whereas the shift toward pure X is primarily associated with reaching higher public states.

This composition of choices provides a more direct comparison with the $(m11)$ benchmark than the mean-$p(X)$ results alone. The $(m11)$ equilibrium predicts interior mixing at $k = 0$ followed by pure X at $k = 1$ and $k = 2$. The data display the same qualitative movement: interior probabilities are most prevalent at $k = 0$, while pure-X choices become increasingly dominant at higher states. The correspondence is not exact, however, because mixing remains present at $k = 1$ and $k = 2$. Thus, the evidence is consistent with movement toward the $(m11)$ state-contingent pattern, but does not indicate exact play of that equilibrium.

Result 5: Mixed behavior declines with both the public state and learning over repeated play. Higher states are associated with a shift toward pure X, while the learning effect primarily appears as a reduction in mixing.

4.6 Baseline comparison

A natural question is what would happen in a standard repeated stag-hunt setting without the public state or the last-round penalty rule. To provide a descriptive benchmark, I conducted two additional $n = 4$ baseline sessions with the same stage-game payoffs and random-termination procedure, but without the public counter or the last-round payoff penalty.[16] The treatment and baseline samples are similar in observable characteristics, although the small number of baseline sessions limits the informativeness of formal balance comparisons.[17]

Table 13 reports the corresponding descriptive outcome comparisons. When $u_Y = 0.15$, both conditions exhibit high rates of X choice and coordination. Mean $p(X)$ is 0.906 in the treatment and 0.967 in the baseline, while the frequency of all-X coordination is 0.719 and 0.854, respectively. Thus, when the safe payoff is low, coordination on X is already close to the upper bound in the absence of the treatment mechanism. The comparison is markedly different when $u_Y = 0.85$. Mean $p(X)$ is 0.577 in the treatment compared with 0.367 in the baseline, a difference of 0.210. Similarly, pure X choices are 18.7 percentage points more frequent and all-X coordination is 16.4 percentage points more frequent in the treatment.

[16] Appendix C1 includes the Experimental Instructions and Interfaces for both $n = 2$ and $n = 4$ baseline sessions, but only $n = 4$ baseline sessions were conducted due to budget constraints.

[17] See Table B1.3 in Appendix B1.

In contrast, the frequency of mixed choices is nearly identical across the two conditions. These suggest that the treatment-baseline difference under high u_Y primarily reflects greater willingness to choose X rather than a change in the prevalence of mixed behavior.

Table 13: Descriptive Comparison of $n = 4$ Treatment and Baseline

	Low u_Y			High u_Y		
Outcome	Treatment	Baseline	Difference	Treatment	Baseline	Difference
Mean $p(X)$	0.906	0.967	-0.061	0.577	0.367	0.210
Pure X: $p(X) = 1$	0.780	0.829	-0.049	0.475	0.288	0.187
Mixed: $0 < p(X) < 1$	0.188	0.168	0.020	0.210	0.203	0.007
all-X coordination	0.719	0.854	-0.135	0.300	0.136	0.164
Sessions	6	2		6	2	

Notes: All entries are descriptive. Outcomes are first averaged within an independent session and then averaged across sessions, so each session receives equal weight. Pure and mixed choices are measured at the subject-round level. All-X coordination is measured once per group-round. Difference is treatment minus baseline.

How do these treatment-baseline differences evolve within and across matches? Figure 11 plots mean $p(X)$ by round within a match. When $u_Y = 0.15$, X choice remains high throughout the match in both conditions, consistent with the ceiling-level behavior reported in Table 13. When $u_Y = 0.85$, however, the two conditions diverge sharply after the first round. In the baseline, $p(X)$ falls rapidly and remains around 0.3 in later rounds, whereas the decline is substantially smaller in the treatment. The aggregate difference between treatment and baseline under high u_Y therefore appears to emerge primarily through different within-match dynamics.

Figure 12 instead plots behavior across match numbers. The across-match pattern is less systematic. Under low u_Y, both conditions remain close to the upper bound throughout the experiment. Under high u_Y, $p(X)$ is generally higher in the treatment than in the baseline, although both series fluctuate across matches. Compared with the pronounced pattern within matches, there is little evidence of a smooth treatment-baseline divergence over repeated matches.

The preceding figures describe how the treatment-baseline difference evolves. To examine more directly where this difference arises in the coordination process, Figure 13 compares behavior by recent coordination history. For the baseline sessions, I construct an "as-if" state using the same transition rule as in the treatment: each match begins at $k = 0$, the state increases following an all-X round up to $k = 2$, and otherwise resets to zero. This constructed state was not observed by baseline subjects and is used only to summarize recent coordination history. When $u_Y = 0.15$, $p(X)$ is high in both conditions at every state. When $u_Y = 0.85$, however, the treatment-baseline difference is concentrated at $k = 0$: the

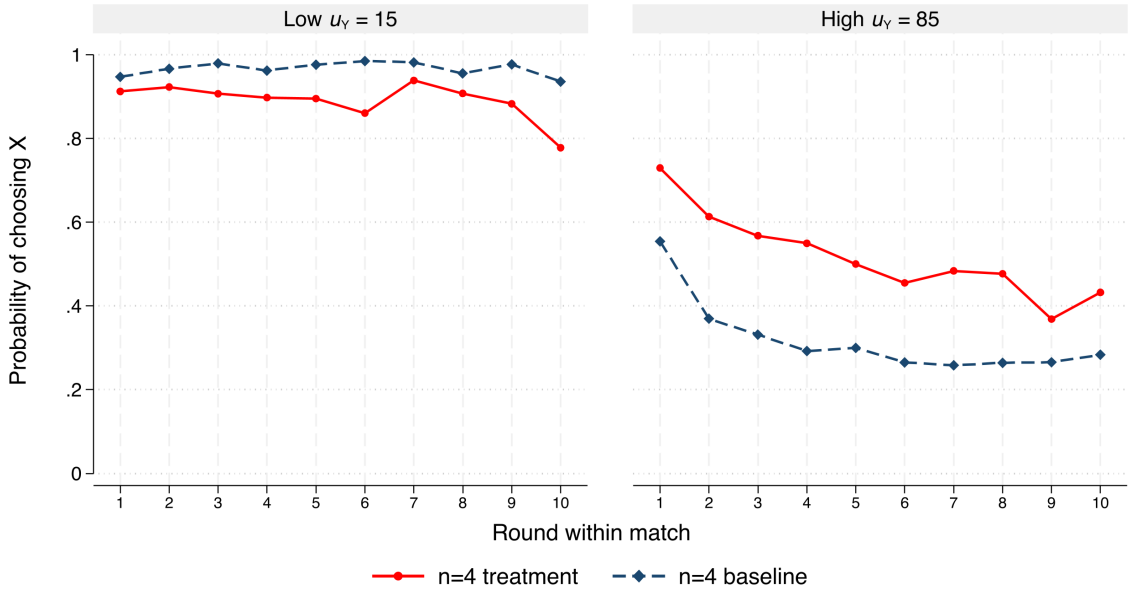


Figure 11: Comparison of $p(X)$ by Round between Treatment and Baseline Sessions

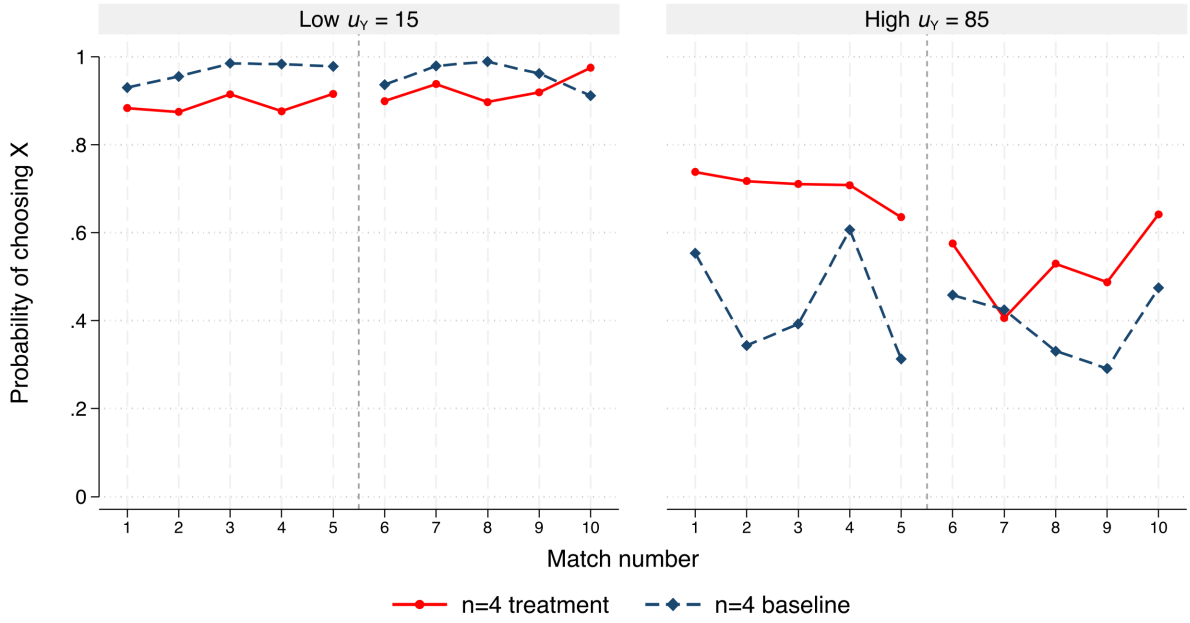


Figure 12: Comparison of $p(X)$ by Match between Treatment and Baseline Sessions

mean $p(X)$ is 0.48 in the treatment compared with 0.28 in the baseline. At $k = 1$ and $k = 2$, $p(X)$ is above 0.9 in both conditions. Thus, the descriptive difference under high u_Y appears primarily when groups have not yet established recent all-X coordination; conditional on

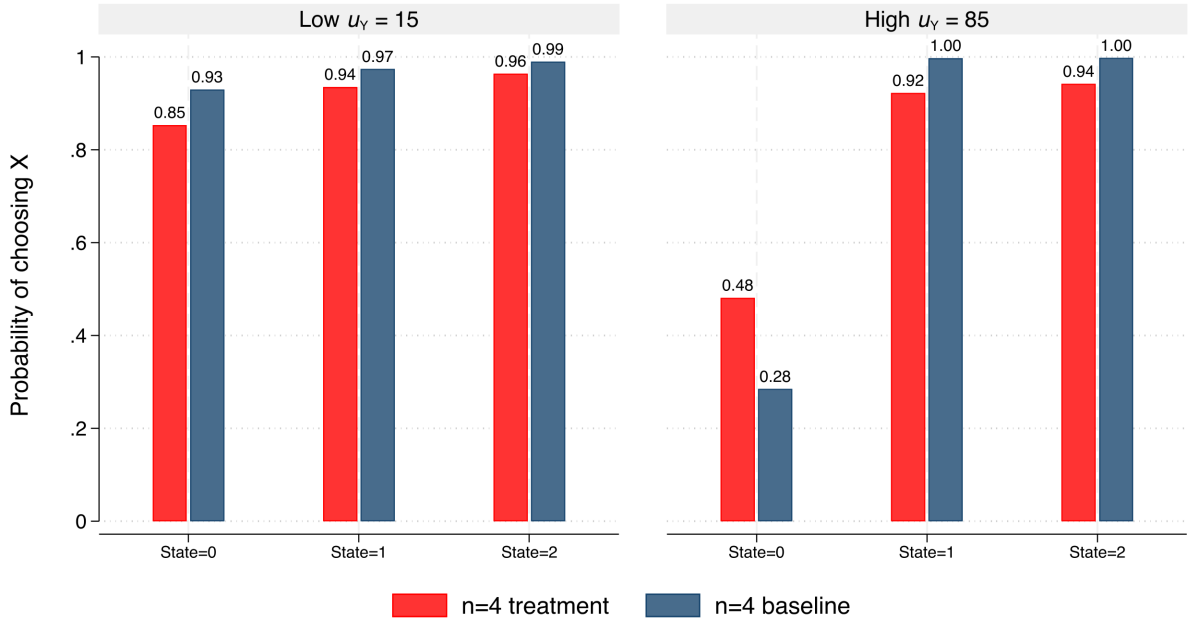


Figure 13: Comparison of $p(X)$ by State between Treatment and Baseline Sessions

having recently coordinated on all-X, behavior is highly persistent in both conditions.

Result 6: Relative to the baseline, the treatment is associated with greater X choice when the safe payoff is high, with the difference concentrated at $k = 0$. When the safe payoff is low, X choice is near the upper bound in both conditions.

4.7 Self-Reported Strategies

The post-experiment survey asked participants about the information they used when making decisions, the consistency of their strategies across matches, and whether experience in earlier matches affected their later strategies. Note that these self-reports are presented only as descriptive evidence and are not used for formal hypothesis testing.

As shown in Table 14, participants generally reported paying attention to past group outcomes rather than relying only on the current public state. In the treatment sessions, 37.8% reported using previous group outcomes only and another 41.5% reported using both previous outcomes and the current counter; only 20.7% reported relying on the counter alone. Similarly, 75.0% of baseline participants reported that their within-match decisions depended on previous group outcomes. Strategies also exhibit some reported persistence across repeated matches. In the treatment sessions, 59.8% of participants reported using the same strategy across matches within both five-match blocks, while 59.1% reported that

Table 14: Self-Reported Strategies in the Post-Experiment Survey

Response	Treatment ($N = 164$)	Baseline ($N = 32$)
Within-match decisions based on the current counter and/or previous group outcomes		
Current counter only	34 (20.7%)	—
Group outcomes in previous rounds only	62 (37.8%)	—
Both current counter and previous outcomes	68 (41.5%)	—
Within-match decisions based on previous group outcomes		
Yes	—	24 (75.0%)
No	—	8 (25.0%)
Strategy consistency within five-match blocks		
Same in both blocks	98 (59.8%)	14 (43.8%)
Same in Matches 1-5 only	25 (15.2%)	11 (34.4%)
Same in Matches 6-10 only	24 (14.6%)	6 (18.8%)
Different in both blocks	17 (10.4%)	1 (3.1%)
Strategies in Matches 6-10 influenced by Matches 1-5		
Yes	97 (59.1%)	16 (50.0%)
No	67 (40.9%)	16 (50.0%)

Notes: Entries are participant counts, with column percentages in parentheses. The first question differs across conditions because baseline sessions did not display the public counter.

their strategies in Matches 6-10 were influenced by their experience in Matches 1-5. The corresponding share for the latter question is 50.0% in the baseline sessions. Taken together, the survey responses suggest that subjects consciously attended to both the current decision environment and their prior coordination experience, complementing the behavioral evidence on state dependence, history dependence, and learning reported above.

5 Discussion and Conclusion

This paper shows that coordination in a dynamic stag hunt depends systematically on accumulated progress. As groups move toward the protection threshold, subjects assign higher probabilities to the risky action and coordination becomes more persistent. The effects of the safe-option payoff and, in strategically difficult environments, group size are also consistent with the comparative statics of the dynamic basin. At the same time, treatment-specific history dependence and incomplete convergence to pure X decisions show that the public state does not fully summarize behavior.

The first main finding is that the public state is strongly associated with coordination behavior (Results 1 and 4). In the pooled sample, the probability assigned to X increases as the group moves toward the threshold, with the largest increase occurring between $k = 0$ and $k = 1$. The stronger prediction that the first state increment exceeds the second receives suggestive support in the pooled analysis, although the difference is not statistically significant at the 5% level. The treatment-specific estimates also reveal substantial heterogeneity. Once groups reach the threshold, however, coordination is highly persistent: most observable post-threshold paths remain coordinated through the end of the match. Together, these patterns are consistent with the central idea of the dynamic-basin framework that coordination becomes more robust as accumulated progress increases. Because the public state is generated by previous group behavior rather than randomly assigned, the estimated state gradients should be interpreted as conditional behavioral associations rather than causal effects of the state itself.

The second main finding concerns state- versus history-dependent behavior (Result 2). In the pooled sample, I do not find a statistically significant difference between behavior at the initial $k = 0$ state and behavior after the state returns to $k = 0$ following a coordination failure. The treatment-specific estimates, however, contain significant departures in opposite directions in the 2l and 4h treatments. Thus, the current public state appears to provide an important summary of the strategic environment, but it does not fully describe behavior under all treatment conditions. One possible interpretation is that a coordination failure changes subjects' beliefs about their current partners even though the objective continuation problem has returned to the same state. The initial-versus-reset comparison should nevertheless be interpreted cautiously because the initial observation occurs in the first round of a match whereas a post-reset observation necessarily occurs later in the match.

The third main finding concerns the comparative statics of strategic uncertainty (Results 3a and 3b). Lowering the safe-option payoff substantially increases the probability assigned to X for both group sizes in the primary within-subject specifications. The effect of group size

is more heterogeneous. When the safe-option payoff is high, groups of two assign substantially higher probabilities to X than groups of four, whereas little group-size difference is observed when the safe-option payoff is low and coordination is already near the upper bound. The latter environment already produces very high levels of X choice, leaving relatively little scope for group size to further increase coordination. More generally, the results suggest that the comparative statics of strategic uncertainty are most visible when the safe alternative is sufficiently attractive that coordination is not already near the upper bound.

The fourth main finding comes from the probability-elicitation design, which reveals behavior that would be hidden by binary action data (Result 5). Interior choice probabilities are common, particularly at the initial state, demonstrating that observed heterogeneity in actions cannot be attributed exclusively to heterogeneity in pure decisions. Mixing becomes substantially less frequent at higher public states, while pure-X choices become increasingly prevalent. Mixing also declines with experience over repeated matches, whereas the corresponding within-block trend in pure-X behavior is small after conditioning on the public state. These patterns suggest two distinct forms of convergence: movement toward higher states is associated primarily with a shift toward pure X, while repeated experience reduces subjects' use of interior probabilities. Because the model admits multiple pure and mixed MPEs, the existence of interior choices alone does not identify selection of a particular mixed equilibrium; the comparison with the $(m11)$ benchmark instead provides a way to assess how closely observed state-contingent behavior resembles the mixed equilibrium predicted by the model.

The baseline sessions provide an additional descriptive benchmark for the role of the dynamic environment (Result 6). When the safe-option payoff is low, both treatment and baseline sessions exhibit high levels of X choice, consistent with a ceiling effect. When the safe-option payoff is high, by contrast, the treatment sessions exhibit greater X choice and all-X coordination, and the difference develops primarily over rounds within a match rather than gradually across matches. This pattern is consistent with the dynamic incentives becoming most important when coordination is otherwise difficult to sustain. The baseline comparison should not, however, be interpreted as isolating a single mechanism: the baseline jointly removes the public counter and the last-round payoff penalty, and only two baseline sessions were conducted. I therefore treat these results as a robustness benchmark rather than as a separate causal treatment comparison.

The experiment is deliberately stylized. The motivating examples, including disease elimination, the adoption of new practices, and minority-culture preservation, are applications in which collective success may require sustained participation and accumulated progress, and the same strategic structure applies more broadly whenever coordinated activity must be

maintained long enough to become resilient to an outside disruption. Real applications involve richer institutions, heterogeneous preferences, communication, and longer time horizons than those represented in the laboratory. The contribution of the experiment is therefore not to reproduce a particular real-world setting, but to isolate a dynamic coordination problem in which current incentives depend on accumulated collective progress.

Overall, the results show that equilibrium selection in a dynamic stag hunt cannot be summarized by the underlying one-shot payoff matrix alone. Coordination behavior varies systematically with accumulated progress, with the attractiveness of the safe option, and, in strategically difficult environments, with the number of agents whose actions must be coordinated. Individual mixing is also quantitatively important at early states but gives way increasingly to pure coordination as groups approach protection. These findings suggest that a state-dependent dynamic basin of attraction provides a useful framework for organizing strategic uncertainty in coordination problems where successful collective action today changes the environment in which coordination takes place tomorrow.

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